Model Categories - Self Study

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Abstract

This text contains my notes on model category theory. These notes were synthesized from a combination of Mark Hovey's *Model Categories*, and Emily Riehl's *Categorical Homotopy Theory*. The sections of my notes on Riehl's text can also be found in my full notes for her text; I made these notes since I am using both texts simultaneously, so it felt wrong to only include these notes in my notes on her book.

I wrote these notes during the summer between my Sophomore and Junior years. My goal was to obtain rigorous understanding of basic model category theory, and to explore Joyal's model structure on sSet, such that I could begin exploring quasi-categories. We also explore Quillen's model structure for $s\mathcal{S}\mathrm{et}$ as a precursor to that of Joyal, as well as the model structures for $\mathcal{C}\mathrm{h}(\mathcal{A})$, $\mathcal{T}\mathrm{op}$, and $\mathcal{M}\mathrm{od}_R$.

Contents

Chapter 1

Model Categories (Hovey 1)

We begin with an initial exploration of model categories following Hovey's more classical perspective, which builds atop that of Quillen. What Quillen originally described as *closed* model categories have been determined to be the more important class of examples, so they have come to be known as model categories in the modern literature. The goal is to provide for a homotopical category some extra structure, in the form of specified classes of morphisms known as fibrations and cofibrations, to make the homotopy category better behaved and less difficult to work with.

Conventions vary mildly on model categorical definitions. Hovey states that we will require functorial factorizations be part of the structure of a model category, which more classical texts did not. I do not yet know Riehl's perspective on this, but either way this distinction, and others like it, are immaterial since in practice all relevant model categories satisfy all definitions.

1.1 The definition of a model category

Given a category $\mathcal{C}$, define by $\operatorname{Map}\mathcal{C}$ the category whose objects are morphisms in $\mathcal{C}$, and whose morphisms are commutative squares; meaning for objects $f : a \to b$ and $g : c \to d$, a morphism $f \to g$ is a pair (h, k) such that

$$\begin{tikzcd} a \arrow[r, "h"] \arrow[d, "f"'] & c \arrow[d, "g"] \\ b \arrow[r, "k"'] & d \end{tikzcd}$$

commutes. One also notes that this is equivalent to the comma category $\operatorname{Map}\mathcal{C} = (\mathcal{C} \downarrow \mathcal{C})$. The operations of domain and codomain promote to functors $\operatorname{dom}, \operatorname{cod} : \operatorname{Map}\mathcal{C} \to \mathcal{C}$, defined by $f \mapsto \operatorname{dom} f$ and $(h, k) \mapsto h$; this works because $h : \operatorname{dom} f \to \operatorname{dom} g$. Codomain is defined similarly. The fact that this is functorial is a subtle yet important nuance in the following definition:

Definition 1.1.1 (Retract/Functorial Factorization)**.** Let $\mathcal{C}$ be a category. A map $f : a \to b$ in $\mathcal{C}$ is a *retract* of a map $g : c \to d$ in $\mathcal{C}$ if f is a retract of g as objects of $\operatorname{Map}\mathcal{C}$. More precisely, f is a retract of g if there is a diagram

$$\begin{tikzcd} a \arrow[r] \arrow[d, "f"'] \arrow[rr, bend left, "1"] & c \arrow[r] \arrow[d, "g"'] & a \arrow[d, "f"] \\ b \arrow[r] \arrow[rr, bend right, "1"'] & d \arrow[r] & b \end{tikzcd}$$

A *functorial factorization* is a pair (α, β) of endofunctors $\operatorname{Map} \mathcal{C} \to \operatorname{Map} \mathcal{C}$ such that $f = \beta(f) \circ \alpha(f)$ for all $f \in \operatorname{Map} \mathcal{C}$, and also such that

- $\operatorname{dom} \circ \alpha = \operatorname{dom}$;
- $\operatorname{cod} \circ \alpha = \operatorname{dom} \circ \beta$;
- $\operatorname{cod} \circ \beta = \operatorname{cod}$.

♡

Remark 1.1.2. Further commentary on the definition of a functorial factorization is beneficial. Note first that the composition $\beta(f) \circ \alpha(f)$ is composition in $\mathcal{C}$, since f is an object of $\operatorname{Map} \mathcal{C}$. We thus obtain the following remarks from this definition:

1. Every arrow f admits a factorization into two components, denoted as $\beta(f)$ and $\alpha(f)$;
2. If one only considers the action of dom and cod on objects, the information in the three bullet points is not anything more powerful than the condition that $\beta(f) \circ \alpha(f)$ be well-defined. However, when once considers the action on morphisms, we obtain that for a morphism $(h, k) : f \to g$ in $\operatorname{Map} \mathcal{C}$,

$$\alpha(k) = (\operatorname{cod} \circ \alpha)(h, k) = (\operatorname{dom} \circ \beta)(h, k) = \beta(h).$$

   Here I use a minor abuse of notation and denote by $\alpha(k)$ the arrow $(\operatorname{cod} \circ \alpha)(h, k)$that replaces k in the square diagram, and similarly for $\beta(h)$. We also obtain that $\alpha(h) = h$ and $\beta(k) = k$. Thus the diagrams

$$\begin{tikzcd} a \arrow[r, "h"] \arrow[d, "\alpha(f)"'] & c \arrow[d, "\alpha(g)"'] \\ b \arrow[r, "\alpha(k)"'] & d \end{tikzcd} \qquad \begin{tikzcd} a \arrow[r, "\beta(h)"] \arrow[d, "\beta(f)"] & c \arrow[d, "\beta(g)"] \\ b \arrow[r, "k"'] & d \end{tikzcd}$$

   glue together to form the diagram

$$\begin{tikzcd} a \arrow[r, "h"] \arrow[d, "\alpha(f)"] \arrow[dd, bend right, "f"'] & b \arrow[d, "\alpha(g)"'] \arrow[dd, bend left, "g"] \\ e_f \arrow[r, "\alpha(k)=\beta(f)"] \arrow[d, "\beta(f)"] & e_g \arrow[d, "\beta(g)"'] \\ b \arrow[r, "k"'] & d \end{tikzcd}$$

3. By the above diagram, functorial factorizatrions preserve retracts;
4. Note also that if

$$f \xrightarrow{\phi} g \xrightarrow{\psi} f$$

   is a retract, then $\psi \circ \phi = 1$, and thus ϕ is mono and ψ is epi.

Also note that the definition as given in Hovey does not make precise that domain and codomain should be viewed as functors; this is clarified in an errata that can be found on the nLab.

Definition 1.1.3 (Lifting Property)**.** Let $i : a \to b$ and $p : x \to y$ be maps in $\mathcal{C}$. Then i has the *left lifting property with respect to* p and p has the *right lifting property with respect to* i if, for every commutative diagram of solid arrows

$$\begin{tikzcd} a \arrow[r, "f"] \arrow[d, "i"'] & x \arrow[d, "p"] \\ b \arrow[r, "g"'] \arrow[ur, dotted, "h"] & y \end{tikzcd}$$

there exists a dotted arrow as indicated rendering the diagram commutative. ♡

Remark 1.1.4. Note that the diagram of solid arrows is the datum of a morphism $(f, g) : i \to p$ in $\operatorname{Map}\mathcal{C}$; a lift h can then be viewed as an arrow in $\mathcal{C}$ such that $(f, h) : i \to 1_x$ and $(h, g) : 1_b \to p$.

Definition 1.1.5 (Model Structure)**.** A *model structure* on a category $\mathcal{C}$ is three *subcategories* of $\mathcal{C}$ called the weak equivalences, fibrations, and cofibrations, together with two functorial factorizations (α, β) and (γ, δ) satisfying the following properties:

1. (2-out-of-3) If f, g are composable arrows in $\mathcal{C}$ such that two of f, g, and gf are weak equivalences, then so is the third;
2. (Retracts) If f, g are morphisms in $\mathcal{C}$ such that f is a retract of g and g is either weak equivalence, fibration, or cofibration, then so is f;
3. (Lifting) A map is an *trivial cofibration* if it is both a cofibration and a weak equivalence, and the same for *trivial fibrations*. Then trivial cofibrations have the left lifitng property with respect to fibrations, and cofibrations have the left lifting property with respect to trivial fibrations;
4. (Factorization) For any morphism f, $\alpha(f)$ is a cofibration, $\beta(f)$ is a trivial fibration, $\gamma(f)$ is a trivial cofibration, and $\delta(f)$ is a fibration.

♡

Definition 1.1.6 (Model Category)**.** A *model category* $\mathcal{C}$ is a complete and cocomplete category with a specified model structure. ♡

Remark 1.1.7. For the naive reader, note that referring to maps as *weak equivalences*, *fibrations*, or *cofibrations* simply means that those maps are in the relevant subcategory of $\mathcal{C}$.

Of course, the intuition for weak equivalences is most easily viewed via homotopy equivalences. Given homotopy equivalences f and g, one immediately has that gf is a homotopy equivalence; conversely given gf and f homotopy equivalences, one may construct a homotopy inverse for g making g a homotopy equivalence, and the same for f. As we well know, the idea of formally inverting a specified class of morphisms is very prominent in mathematics, with the inversion of homotopy equivalences in $\mathcal{T}\mathrm{op}$ closely tied to quotienting by homotopy, and the inversion of quasi-isomorphisms in $\mathcal{C}\mathrm{h}(\mathcal{A})$ giving rise to the derived category $\mathcal{D}(\mathcal{A})$, a central topic in homological algebra. Due to my lacking background in algebraic topology, I lack the natural intuition for what fibrations and cofibrations should be, which gives me an opportunity to develop a uniquely categorical intuition before I end up reading on algebraic topology.

With the 2-out-of-3 property thoroughly motivated, we give some more detailed explanation for the other properties.

- (Retracts) This property simply says that fibrations, cofibrations, and weak equivalences are closed under retracts.

- (Lifting) It is not detailed thoroughly in this text, but for a collection $\mathcal{L}$ of maps to have the left lifting property against another collection $\mathcal{R}$ of maps, we must have that *every* map in $\mathcal{L}$ has the left lifting property with respect to *every* map in $\mathcal{R}$.
- (Factorization) This property says that each map admits a factorization into a trivial cofibration and a fibration, and a cofibration and a trivial fibration, in the sense of definition 1.1.1 and remark 1.1.2.

Remark 1.1.8. Emily Riehl in *Categorical Homotopy Theory* requires that model categories satisfy a stronger 2-of-6 property, instead of the 2-out-of-3 property. It's proven in her text that in the case of model categories, these properties become equivalent; I will prove this when I read her discussion of model categories. The more important distinction she makes in her text is that the class of weak equivalences needs to be *wide*, meaning it contains all objects. This is a simple way of saying it contains all identities, and thus by a very simple proof it contains all isomorphisms. Hovey does not specify this, but it would still make sense for a model category's class of weak equivalences to contain all isomorphisms. This is reconsiled by a result we will not see until later. The lifting axiom will later be shown to *characterize* trivial (co)fibrations, meaning we will show maps are trivial cofibrations *if and only if* they have the left lifting property with respect to fibrations, and similarly for the dual statement. Note that isomorphisms admit a lift for any diagram: let f and g be inverses. Then

$$\begin{tikzcd}
x \arrow[r, "h"] \arrow[d, "f"'] & a \arrow[d, "p"] \\
y \arrow[r, "k"'] \arrow[ur, dotted, "x" description] & b
\end{tikzcd}$$

commutes, since $hgf = h$ gives the commutivity of the top triangle, and since the outer square commutes by hypothesis, $kf = ph \implies k = phg$ as required for the bottom triangle. Thus f has the left lifting property against all arrows; in particular fibrations. Thus f, as will be shown in lemma 1.1.18, must be a trivial cofibration, and thus it is by definition a weak equivalence.

We assume small limits and colimits exist largely for convenience, since most model categories in practice admit all small limits. This differs from Quillen's original requirement that only finite limits exist. We also assume the stronger convention that factorizations be functorial, since once again they generally are in practice. We also assume the functorial factorizations are *part of the structure*, which is a subtle difference and ensures that some maps will be natural going forward. We always leave the model structure implicit and refer to a model category $\mathcal{C}$, because oh my god can you imagine having to write out like $(\mathcal{C}, (\alpha, \beta), (\gamma, \delta), \mathcal{F}, \mathcal{Q}, \mathcal{W})$ every time you wanted to refer to a model category? That would be so aids lmfao

Example 1.1.9. Any complete and cocomplete category $\mathcal{C}$ can be given three different model structures by choosing one of the subcategories to be isomorphisms, and the other two subcategories to be all of $\mathcal{C}$. The only nontrivial things to be shown are that

1. retracts of isomorphisms are isomorphisms;
2. the lifting axiom holds;
3. functorial factorizations admit a definition.

Proving (1) is not incredibly difficult. Let $g : x \to y$ be an isomorphism, and let $f : a \to b$ be a retract of g:

$$\begin{tikzcd}
a \arrow[r, "h"] \arrow[d, "f"'] & x \arrow[r, "h'"] \arrow[d, "g"'] & a \arrow[d, "f"] \\
b \arrow[r, "k"'] & y \arrow[r, "k'"'] & b
\end{tikzcd}$$

By commutivity of the diagram, we have that $k'g = fh'$, and thus $k' = fh'g^{-1}$. We thus construct the diagram

$$\begin{tikzcd}
a \arrow[r, "h"] \arrow[d, "f"'] & x \arrow[r, "h'"] \arrow[d, "g"'] & a \arrow[d, "f"] \\
b \arrow[r, "k"'] \arrow[dr, "g^{-1}k"'] & y \arrow[r, "k'"'] \arrow[d, "g^{-1}"'] & b \\
& x \arrow[ur, "fh'"'] &
\end{tikzcd}$$

Since the composition along the lower row is the identity, we obtain that $fh'g^{-1}k = 1$, and thus $h'g^{-1}k$ is a right inverse for f. Showing that it is a left inverse is more simple; by commutivity of the left square we have

$$gh = kf \implies h = g^{-1}kf \implies 1 = h'g^{-1}kf,$$

where we have used the fact that $h'h = 1$. Thus, we have constructed a two-sided inverse for f, and f must be an isomorphism.

Now we prove (2) for each case. We will show extremely soon that model categories are dualizable, so we need only show it for one of the cofibrations or fibrations.

1. Start by taking weak equivalences to be precisely isomorphisms. Thus trivial cofibrations and trivial fibrations are precisely isomorphisms, which we showed in remark 1.1.8 to have the left lifting property with respect to all maps; the same proof shows they have the right lifting property as well (with notation as in the problem, the diagonal arrow is gk. Thus trivial (co)fibrations have the left *and* right lifting properties with respect to all maps, and thus (co)fibrations. This proves the statement.

2. Now take cofibrations to be isomorphisms. All arrows are trivial fibrations, and all cofibrations are trivial cofibrations. Since trivial cofibrations are in particular isomorphisms, they have the left lifting property with respect to all maps, and thus fibrations. Note also that all cofibrations are isomorphisms, and thus have the left lifting property with respect to all maps, in particular trivial fibrations.

Now we prove (3). We once again work case-by-case.

1. Take weak equivalences to be isomorphisms. Then one defines $\alpha(f) = f$ and $\beta(f) = 1_{\operatorname{cod} f}$, similarly $\gamma(f) = 1_{\operatorname{dom} f}$ and $\delta(f) = f$. The functorial factorization condition forces the action on a morphism $(h, k) : f \to g$ to satisfy

$$(\operatorname{dom} \circ\alpha)(h, k) = (\operatorname{dom} \circ\gamma)(h, k) = h, \qquad (\operatorname{cod} \circ\beta)(h, k) = (\operatorname{cod} \circ\delta)(h, k) = k.$$

One defines $(\operatorname{cod} \circ\alpha)(h, k) = k$ and $(\operatorname{cod} \circ\gamma)(h, k) = h$, and the corresponding definition for β, δ. This is indeed functorial, since all the factorization (α, β) does is attach a square to the bottom:

$$\begin{tikzcd}
x \arrow[r, "h"] \arrow[d, "f"'] & a \arrow[d, "g"] \\
y \arrow[r, "k"'] \arrow[d, "1"'] & b \arrow[d, "1"] \\
y \arrow[r, "k"'] & b
\end{tikzcd}$$

and similarly for (γ, δ). Note that in this case, we find that α, δ are the identity and β, γ can be described as the identity on the codomain.

2. I will not go into as much detail for this case since the proof is basically the same, but with some things switched around. Also because this stuff is kinda trivial. Let isomorphisms be cofibrations. Since identities are cofibrations, we decompose every map as itself composed with 1; this produces both functorial factorizations, since all cofibrations are trivial and all fibrations are trivial.

Example 1.1.10 (Product Model Structure)**.** Let $\mathcal{C}$ and $\mathcal{D}$ be model categories. One defines the *product model structure* on $\mathcal{C} \times \mathcal{D}$ by taking maps (f, g) to be cofibrations (respectively fibrations, weak equivalences) if and only if both f and g are cofibrations (respectively fibrations, weak equivalences). The verification of all of the axioms is kind of trivial, since everything just decomposes into noting that the axioms hold in both categories, so it holds for the pairs by definition of the product category. The functorial factorizations are defined as the product; for example

$$\alpha_{\mathcal{C}} \times \alpha_{\mathcal{D}} : \operatorname{Map}\mathcal{C} \times \operatorname{Map}\mathcal{D} \to \operatorname{Map}\mathcal{C} \times \operatorname{Map}\mathcal{D},$$

where we note that $\operatorname{Map}\mathcal{C} \times \operatorname{Map}\mathcal{D} \cong \operatorname{Map}(\mathcal{C} \times \mathcal{D})$.

Remark 1.1.11. Model category axioms are self-dual, meaning if $\mathcal{C}$ is a model category, $\mathcal{C}^{\mathrm{op}}$ inherits a corresponding model structure by taking cofibrations in $\mathcal{C}^{\mathrm{op}}$ to be fibrations in $\mathcal{C}$; fibrations in $\mathcal{C}^{\mathrm{op}}$ to be cofibrations in $\mathcal{C}$; and weak equivalences stay the same. The functorial factorizations are inverted: denote by $(-)'$ the corresponding functor for $\mathcal{C}^{\mathrm{op}}$. Then

- $\alpha' = \delta^{\mathrm{op}}$;
- $\beta' = \gamma^{\mathrm{op}}$;
- $\gamma' = \beta^{\mathrm{op}}$;
- $\delta' = \alpha^{\mathrm{op}}$.

We denote by $D\mathcal{C}$ the opposite model category of $\mathcal{C}$ with this model structure, and call it the *dual model category* of $\mathcal{C}$. Note that $D^2\mathcal{C} = \mathcal{C}$. Of course this means we need only prove a statement about cofibrations to prove the dual for fibrations, and vice versa.

Terminology 1.1.12. Since model categories are complete and cocomplete, they have initial and terminal objects. We denote by 0 the initial object, and 1 or $*$ the terminal object. An object $x \in \mathcal{C}$ is *cofibrant* if the arrow $0 \to x$ is a cofibration, and is called *fibrant* if the arrow $x \to *$ is a fibration. A model category is *pointed* when the map $0 \to *$ is an isomorphism.

Construction 1.1.13. Let $\mathcal{C}$ be a model category. Define by $\mathcal{C}_*$ the category $(* \downarrow \mathcal{C})$ under the terminal object. Recall this is the usual definition of a based category, as objects are arrows

$f : * \to x$, which are intuitively though of as selecting a basepoint f for x. Recall morphisms are basepoint-preserving maps, so maps $\varphi : (x, f) \to (y, g)$ such that $g = \varphi \circ f$.

Note the obvious functor $(-)_+ : \mathcal{C} \to \mathcal{C}_*$ given by $x \mapsto x \amalg * =: x_+$, with the obvious basepoint. This is clearly an adjunction $(-)_+ \dashv U$ for $U : \mathcal{C}_* \to \mathcal{C}$ the forgetful functor, and thus $(-)_+$ preserves colimits. Thus, $\mathcal{C}_*$ is cocomplete. One also shows fairly easily that $\mathcal{C}_*$ is complete with limits in $\mathcal{C}_*$ the same as the underlying object in $\mathcal{C}$. One typically denotes by $x \vee y$ the coproduct in $\mathcal{C}_*$, analogous to the wedge product in $\mathcal{T}\mathrm{op}_*$.

We construct a model structure on $\mathcal{C}_*$ by defining maps f in $\mathcal{C}_*$ to be cofibrations (fibrations, weak equivalences) precisely when the underlying function $U(f)$ is a cofibration (fibration, weak equivalence).

Proposition 1.1.14. *With definitions as in construction 1.1.13, $\mathcal{C}_*$ is a model category.*

Proof. This is really not a hard proof. Composition preserves the underlying function, so the 2-out-of-3 property holds. To show the subcategories are closed under retracts, we need only show that the underlying objects also form a retract. Indeed, this follows by functorality of the forgetful functor, so this follows. We will now consider the factorization axiom, and will come back to lifting.

We may lift the functorial factorizations to $\mathcal{C}_*$ mostly for free: the functorial factorization condition guarentees the domain of $\alpha(h, k)$ is h, the codomain of $\beta(h, k)$ is k, and given a morphism $f \to g$ in $\operatorname{Map} \mathcal{C}_*$ we would already have that these are basepoint preserving. The only work that needs to be done is to come up with a basepoint for e_f and e_g, the midpoint of $\beta(f) \circ \alpha(f)$, such that the maps are basepoint preserving. The correct definition manifests from the following diagrams: an arrow $f : (x, u) \to (y, v)$ in $\mathcal{C}_*$ can be viewed as a diagram

$$\begin{tikzcd} & x \arrow[dd, "f"] \\ * \arrow[ur, "u"] \arrow[dr, "v"'] & \\ & y \end{tikzcd}$$

in $\mathcal{C}$. Since α and β are a factorization in $\mathcal{C}$, we obtain the diagram

$$\begin{tikzcd} & x \arrow[dd, "f"] \arrow[dr, "\alpha(f)"] & \\ * \arrow[ur, "u"] \arrow[dr, "v"'] & & e_f \arrow[dl, "\beta(f)"] \\ & y & \end{tikzcd}$$

in $\mathcal{C}$. Hence we may define the basepoint of e_f as $\alpha(f) \circ u : * \to e_f$, and it holds manifestly that the relevant diagram in $\mathcal{C}_*$ commutes, since it commutes in $\mathcal{C}$. All one has to do in order to realize this is to draw the diagram with and without the endpoints, look at it for 10 seconds, say "oh thats obvious", and move on. The fact that these factor into cofibrations and trivial fibrations is immediate by the definition of the model structure. The definitions of γ and δ follow in the same way.

Finally, we concern ourselves with the lifting properties. Suppose we have a lifting problem

$$\begin{tikzcd} (x,u) \arrow[r, "\mu"] \arrow[d, "p"'] & (a,w) \arrow[d, "i"] \\ (y,v) \arrow[r, "\nu"'] & (b,z) \end{tikzcd}$$

in $\mathcal{C}_*$, with p a cofibration and i a trivial fibration. Functoriality of U gives the lifting problem

$$\begin{tikzcd} x \arrow[r, "\mu"] \arrow[d, "p"'] & a \arrow[d, "i"] \\ y \arrow[r, "\nu"'] \arrow[ur, dotted, "h"] & b \end{tikzcd}$$

with the solution h as indicated, since $\mathcal{C}$ is a model category. Since the diagram without h commutes in $\mathcal{C}_*$, the outer arrows are basepoint preserving, meaning the diagram

$$\begin{tikzcd} & x \arrow[rr, "\mu"'] \arrow[dd, "p"] & & a \arrow[dd, "i"] \\ * \arrow[ur, "u"'] \arrow[dr, "v"] \arrow[urrr, bend left, "w"] \arrow[drrr, bend right, "z"'] & & & \\ & y \arrow[rr, "\nu"] \arrow[uurr, dotted, "h"] & & b \end{tikzcd}$$

must also commute by basepoint preservation. To show that h solves the lifting problem in $\mathcal{C}_*$, it suffices to show it preserves basepoints, since we already know the diagram will commute. Thus, we must show $hv = w$. This follows immediately from commutivity of the diagram. The dual statements for trivial cofibrations and fibrations once again follow from remark 1.1.11. Thus, $\mathcal{C}_*$ is a model category. ■

Remark 1.1.15. The proof of proposition 1.1.14 did not rely on the universality of $*$ at all, meaning one can generate for any object $a \in \mathcal{C}$ a model category structure on $\mathcal{C}_a := (a \downarrow \mathcal{C})$ via the same assignments. Dualizing this gives us that the category $(\mathcal{C} \downarrow a)$ is also a model category.

Remark 1.1.16. Note that for any object $x \in \mathcal{C}$, the unique arrow $f : 0 \to x$ decomposes as

$$0 \xrightarrow{\alpha(f)} Q(x) \xrightarrow{\beta(f)} x,$$

where the reason for calling the in-between object $Q(x)$ will become clear soon. By definition 1.1.5, the arrow $\alpha(f)$ is a cofibration, so we have that the object $Q(x)$ is cofibrant. We thus obtain that the functor $Q = \operatorname{cod} \circ \alpha$ maps objects x to *cofibrant objects* $Q(x)$. We call Q the *cofibrant replacement functor* of $\mathcal{C}$. We also obtain a natural transformation $q : Q \Rightarrow 1_{\mathcal{C}}$ with components given by $q_x = \beta(0 \to x)$, wherein each component is a trivial fibration. The fact that this is natural comes from the fact that the functorial factorization defines, for each $f : x \to y$, a diagram

$$\begin{tikzcd} 0 \arrow[r, "\alpha(0\to x)"] \arrow[d, equal] & Q(x) \arrow[r, "\beta(0\to x)"] \arrow[d, "\operatorname{cod}\alpha(1,f)"'] & x \arrow[d, "f"] \\ 0 \arrow[r, "\alpha(0\to y)"'] & Q(y) \arrow[r, "\beta(0\to y)"'] & y \end{tikzcd}$$

First note that the outside square commutes since 0 is initial, and thus both inner squares commute since (α, β) is a functorial factorization. Note that $\operatorname{cod}\alpha(1, f) = Q(f)$, so the right square expresses the naturality condition as desired. Dually, we have that $\operatorname{dom} \circ \delta$ is an *fibrant replacement*, usually denoted $R : \mathcal{C} \to \mathcal{C}$, and there is a natural trivial cofibration[1] $r : 1_{\mathcal{C}} \Rightarrow R$ with components given by $r_x = \gamma(0 \to x)$.

Lemma 1.1.17 (The Retract Argument)**.** *Suppose one has the factorization $f = p \circ i$ in $\mathcal{C}$, and suppose that f has the left lifting property with respect to p. Then f is a retract of i. Dually, if f has the right lifting property with respect to i, then f is a retract of p.*

Proof. Define the domains via

$$\begin{tikzcd} x \arrow[r, "i"] \arrow[rr, bend left, "f"] & z \arrow[r, "p"] & y. \end{tikzcd}$$

Suppose f has the left lifting property with respect to p. Note that since $f = pi$, the following diagram of solid arrows

$$\begin{tikzcd} x \arrow[r, "i"] \arrow[d, "f"'] & z \arrow[d, "p"] \\ y \arrow[r, equal] \arrow[ur, dotted, "r"] & y \end{tikzcd}$$

commutes, and thus there is a dotted arrow r as indicated rendering the diagram commutative. This expresses the equality $i = rf$, and thus the diagram

$$\begin{tikzcd} x \arrow[r, equal] \arrow[d, "f"'] & x \arrow[r, equal] \arrow[d, "i"'] & x \arrow[d, "f"] \\ y \arrow[r, "r"'] & z \arrow[r, "p"'] & y \end{tikzcd}$$

commutes, since the equality expressed by the square on the right is simply $f = pi$. Note also by the diagram that $pr = 1$, and thus f is a retract of i. The other argument follows by duality. ■

Lemma 1.1.18. *Let $\mathcal{C}$ be a model category. Then a map is a (trivial) cofibration* if and only if *it has the left lifting property with respect to all fibrations (trivial fibraitons). Dually, a map is a fibraiton (a trivial fibration)* if and only if *it ha the right lifting property with respect to all trivial cofibrations (cofibrations).*

Proof. The model category axioms guarentee that (trivial) cofibrations lift against fibrations (trivial fibrations), so we need only show the converse. Let f have the left lifting property with

[1]The terminology "natural [class of map]" means there is a natural transformation, and each component belongs to the specified type of map. These are also called "pointwise [class of map] natural transformations".

respect to all trivial fibrations. Note that $f = \beta(f) \circ \alpha(f)$, and $\beta(f)$ is a trivial fibration. By assumption, f thus has the left lifting property with respect to $\beta(f)$, and by lemma 1.1.17 f is a retract of $\alpha(f)$. Since $\alpha(f)$ is a cofibration and cofibrations are closed under retracts, we have that f is a cofibration.

Next, let f have the left lifting property with respect to all fibrations, and note that $f = \delta(f) \circ \gamma(f)$, where $\delta(f)$ is a fibration. By assumption, f has the left lifting property with respect to $\delta(f)$, and by lemma 1.1.17 f is a retract of $\gamma(f)$. Since $\gamma(f)$ is a trivial cofibration, it is a weak equivalence and a cofibration, both of which are closed under retracts. Therefore, f is a trivial cofibration. The other statements follow by duality. ■

Remark 1.1.19. This suffices to prove remark 1.1.8, in particular that all isomorphisms are weak equivalences. In particular, isomorphisms are trivial cofibrations *and* trivial fibrations, since they have the retract argument against all maps (again, see remark 1.1.8).

Corollary 1.1.20. *Let $\mathcal{C}$ be a model category. Then cofibrations (trivial cofibrations) are closed under pushouts in the sene that if*

$$\begin{tikzcd} a \arrow[r] \arrow[d, "f"'] & c \arrow[d, "g"] \\ b \arrow[r] & d \end{tikzcd}$$

is a pushout square and f is a cofibration (trivial cofibration), then so is g. Dually, fibrations are closed under pullbacks in the sense that if

$$\begin{tikzcd} a \arrow[r] \arrow[d, "g"'] & c \arrow[d, "f"] \\ b \arrow[r] & d \end{tikzcd}$$

is a pullback and f is a fibration (trivial fibration), then so is g.

Proof. Consider the pushout case. Let

$$\begin{tikzcd} a \arrow[r, "\ell_1"] \arrow[d, "f"'] & c \arrow[d, "g"] \\ b \arrow[r, "k_1"'] & d \end{tikzcd}$$

be the pushout square, and consider a lifting problem

$$\begin{tikzcd} c \arrow[r, "\ell_2"] \arrow[d, "g"'] & x \arrow[d, "h"] \\ d \arrow[r, "k_2"'] & y. \end{tikzcd}$$

Attaching the pushout square gives the diagram

$$\begin{tikzcd} a \arrow[r, "\ell_1"] \arrow[d, "f"'] & c \arrow[r, "\ell_2"] \arrow[d, "g"'] & x \arrow[d, "h"] \\ b \arrow[r, "k_1"'] & d \arrow[r, "k_2"'] & y. \end{tikzcd}$$

If f has the left lifting property with respect to h, then there is a lift

$$\begin{tikzcd} a \arrow[r, "\ell_1"] \arrow[d, "f"'] & c \arrow[r, "\ell_2"] & x \arrow[d, "h"] \\ b \arrow[r, "k_1"'] \arrow[urr, dotted, "r"] & d \arrow[r, "k_2"'] & y. \end{tikzcd}$$

indicated by the dotted line. Commutivity of these diagrams gives the diagram of solid arrows

$$\begin{tikzcd} a \arrow[r, "\ell_1"] \arrow[d, "f"'] & c \arrow[d, "g"] \arrow[ddr, bend left, "\ell_2"] & \\ b \arrow[r, "k_1"'] \arrow[drr, bend right, "r"'] & d \arrow[dr, dashed, "r'"] & \\ & & x \end{tikzcd}$$

and by unversality of pushout, there is a unique dashed arrow r' rendering the diagram commutative. We thus obtain that the diagram

$$\begin{tikzcd} c \arrow[r, "\ell_2"] \arrow[d, "g"'] & x \\ d \arrow[ur, "r'"'] & \end{tikzcd}$$

commutes. To show the full relevant diagram commutes, note that composing in the bottom right of the pushout diagram by h gives a unique arrow $hr'k_1 = hr$, and since $hr = k_1k_2$, we have $hr'k_1 = k_2k_1$. By uniqueness of the arrow hr', we have tht $hr' = k_2$, and thus we may fill in our diagram to obtain the commutative diagram

$$\begin{tikzcd} c \arrow[r, "\ell_2"] \arrow[d, "g"'] & x \arrow[d, "h"] \\ d \arrow[ur, "r'"'] \arrow[r, "k_2"'] & y \end{tikzcd}$$

Thus, if f has the left lifting property with respect to any arrow h, then so does g. The statement of the corollary then follows by remark 1.1.19. ■

Lemma 1.1.21 (Ken Brown's Lemma)**.** *Let $\mathcal{C}$ be a model category and $\mathcal{D}$ a homotopic category*[2]*. Let*

[2]A category which has a specified class of weak equivalences which satisfy the 2-out-of-3 axiom. This differs from Riehl's definition, but in particular the 2-out-of-6 axiom implies the 2-out-of-3 axiom, so Ken Brown's lemma applies

$F : \mathcal{C} \to \mathcal{D}$ be a functor which takes trivial cofibrations between cofibrant objects to weak equivalences. Then F takes all weak equivalences between cofibrant objects to weak equivalences. Dually, if F takes tivial fibrations between fibrant objects to weak equivalences, then F takes all weak equivalences between fibrant objects to weak equivalences.

Proof. Let $f : a \to b$ is a weak equivalence between cofibrant objects. The map $(f, 1_b) : a \amalg b \to b$ factors via (α, β) into a cofibration $q : a \amalg b \to c$ and a trivial fibration $p : c \to b$. Since 0 is initial,

$$\begin{tikzcd} 0 \arrow[r] \arrow[d] & a \arrow[d] \\ b \arrow[r] & a \amalg b \end{tikzcd}$$

is a pushout. Since a and b are cofibrant objects, corollary 1.1.20 shows that the inclusions $i_a : a \to a \amalg b$ and $i_b : b \to a \amalg b$ are cofibrations. Note that by definition of the coproduct, $pqi_a = f$ and $pqi_b = 1_b$. Since p is also a weak equivalence, since it is a trivial fibration, we have by the 2-out-of-3 axiom that qi_a and qi_b are weak equivalences, and thus trivial cofibrations. In particular, they are trivial cofibrations $b \to c$ and $a \to c$. Note also that the composition

$$0 \longrightarrow a \xrightarrow{i_a} a \amalg b \xrightarrow{q} c$$

is a composition of cofibrations, and thus c is cofibrant. Therefore, qi_a and qi_b are trivial cofibrations between cofibrant objects. By hypothesis, $F(qi_a)$ and $F(qi_b)$ are weak equivalences. Since $F(pqi_b) = F(1_b)$ is also a weak equivalence, by the 2-out-of-3 property $F(p)$ is a weak equivalence. By the 2-out-of-3 property, $F(p \circ q \circ i_1) = F(f)$ is a weak equivalence. ∎

Intuition 1.1.22. Ken Brown's lemma says if a functor preserves trivial cofibrations between cofibrant objects, then we can expand it to all weak equivalences, and vice versa for fibrants.

1.2 The homotopy category

In this section, we follow Quillen's original approach with few modifications to construct the homotopy category $\operatorname{Ho}\mathcal{C}$ by formally inverting all weak equivalences in $\mathcal{C}$. The main result is that the quotient category $\mathcal{C}_{cf}/\sim$ obtained by quotienting cofibrant and fibrant objects by the homotopy relation is equivalent as a category to $\operatorname{Ho}\mathcal{C}$. We will see what this means soon.

Definition 1.2.1 (Homotopy Category)**.** Let $\mathcal{C}$ be a homotopic category with weak equivalences $\mathcal{W}$. The *homotopy category* $\operatorname{Ho}\mathcal{C}$ is given as follows: Take the free category $F(\mathcal{C}, \mathcal{W}^{-1})$ on $\mathcal{C}$ and $\mathcal{W}^{-1}$, where the latter is the formal inversion of $\mathcal{W}$. Objects are objects of $\mathcal{C}$, and morphisms are chains of composable pairs $(f_1, \ldots, f_n)$ where each f_i is either an arrow in $\mathcal{C}$ or the formal reversal w_i^{-1} of an arrow in $\mathcal{W}$. Composition is string concatenation, and the identities are empty strings. Quotient by the relations $1_a = (1_a)$ for all $a \in \mathcal{C}$, $(f, g) = (g \circ f)$ for all composable pairs (f, g), and $1_{\operatorname{dom} w} = (w, w^{-1})$ and $1_{\operatorname{cod} w} = (w^{-1}, w)$ for all $w \in \mathcal{W}$. The resultant category is $\operatorname{Ho}\mathcal{C}$. ♡

Intuition 1.2.2. The way to view these relations is that we may freely remove identities from strings without changing their value, we may freely compose arrows without changing their value, and this composition assigns for each $w \in \mathcal{W}$ an inverse w^{-1}.

to her notion of a homotopic category as well.

Remark 1.2.3. The category $\operatorname{Ho}\mathcal{C}$ is also sometimes denoted by $\mathcal{C}[\mathcal{W}^{-1}]$, since it is the localization of $\mathcal{C}$ at $\mathcal{W}$. Also note that hom-sets may not in general be sets in $\operatorname{Ho}\mathcal{C}$, so we must implicitly pass to a higher universe in general. We will show soon that this is not necessary in the context of model categories. It is clear from the definition that $\operatorname{Ho} D\mathcal{C} = (\operatorname{Ho}\mathcal{C})^{\mathrm{op}}$ if $\mathcal{C}$ is a model category. We can also show that given model categories $\mathcal{C}$ and $\mathcal{D}$, $\operatorname{Ho}(\mathcal{C} \times \mathcal{D}) \cong \operatorname{Ho}\mathcal{C} \times \operatorname{Ho}\mathcal{D}$.

One notes that we should intuitively think of $\operatorname{Ho}\mathcal{C}$ as the "quotient" by the relation that maps in $\mathcal{W}$ be inverted, and thus we would expect it to be characterized by the universal property of quotients; namely that there is a localization functor $\gamma : \mathcal{C} \to \operatorname{Ho}\mathcal{C}$ which is the identity on objects and $f \mapsto [f]$ on morphisms, which should be initial with respect to functors that invert weak equivalences. Indeed, such a functor clearly exists, and the following lemma verifies the universal property.

Lemma 1.2.4. *Suppose $\mathcal{C}$ is any category with a subcategory $\mathcal{W}$.*

1. *If $F : \mathcal{C} \to \mathcal{D}$ is a functor that sends maps of $\mathcal{W}$ to isomorphisms, then there is a unique functor $\operatorname{Ho} F : \operatorname{Ho}\mathcal{C} \to \mathcal{D}$ such that $(\operatorname{Ho} F) \circ \gamma = F$;*
2. *Suppose $\delta : \mathcal{C} \to \mathcal{E}$ is a functor that takes maps of $\mathcal{W}$ to isomorphisms and satisfies the universal property of (1). Then there is a unique isomorphism $F : \operatorname{Ho}\mathcal{C} \to \mathcal{E}$ such that $F\gamma = \delta$;*
3. *The correspondence of part (1) induces an isomorphism of categories between the functor category $\mathcal{D}^{\operatorname{Ho}\mathcal{C}}$ and the subcategory of functors $\mathcal{D}^{\mathcal{C}}$ which invert isomorphisms of $\mathcal{W}$.*

Proof. One could skip most of this proof and simply appeal to the fact that this is a special case of a quotient category, for which we already have these results, but in order to be thorough I work through the proof without appealing to this fact.

Define $\operatorname{Ho} F$ to be F on objects, and $\operatorname{Ho} F(w^{-1}) = F(w)^{-1}$. We also define $\operatorname{Ho} F$ to be F on arrows not in $\mathcal{W}^{-1}$; an inspection of the relations we are quotienting by shows that this is well-defined:

- we quotient by the relation that we may compose arrows. This is an axiom of categories, and thus all functors necessarily identify these arrows simply by the fact that their domain is a category;
- we quotient by the relation that composing by 1 doesn't change the value of a composition. Once again, this is an axiom of categories themselves;
- we quotient by the relation that arrows in $\mathcal{W}$ have inverses, which $\operatorname{Ho} F$ respects.

Functoriality follows since F is functorial: $\operatorname{Ho} F(gf) = F(gf) = F(g)F(f)$ provided $g, f \notin \mathcal{W}^{-1}$; if $u, v \in \mathcal{W}$ then define $w = uv$, and we have $w^{-1} = v^{-1}u^{-1} = (uv)^{-1}$. We obtain

$$\operatorname{Ho} F(v^{-1}u^{-1}) = \operatorname{Ho} F(w^{-1}) = F(w)^{-1};$$

$$\operatorname{Ho} F(v^{-1}) \operatorname{Ho} F(u^{-1}) = F(v)^{-1}F(u)^{-1} = (F(u)F(v))^{-1} = F(uv)^{-1} = F(w)^{-1}.$$

Thus, our definition applies on $\mathcal{W}^{-1}$ as well. A similar proof takes care of the case where only one arrow is in $\mathcal{W}^{-1}$. We then have

$$(\operatorname{Ho} F \circ \gamma)(x) = \operatorname{Ho} F(x) = F(x), \qquad (\operatorname{Ho} F \circ \gamma)(f) = \operatorname{Ho} F(f) = F(f),$$

and thus $\operatorname{Ho} F \circ \gamma = F$. To show uniqueness, let $\operatorname{Ho} F \circ \gamma = F$; we must show the definition of $\operatorname{Ho} F$ is forced. Indeed, $\operatorname{Ho} F$ must be F on objects and arrows, so we only need to show our definition on $\mathcal{W}^{-1}$ is forced. Indeed, this definition is forced by functoriality and basic properties of isomorphisms, since $\operatorname{Ho} F(w^{-1})$ must be the unique two-sided inverse of $\operatorname{Ho} F(w) = F(w)$, and thus $\operatorname{Ho} F(w^{-1}) = F(w)^{-1}$. This shows property (1).

Now for property (2). Let $\delta : \mathcal{C} \to \mathcal{E}$ also be universal with respect to inverting $\mathcal{W}$. This proof follows in the standard way: by part (1) there is a unique functor $F : \operatorname{Ho} \mathcal{C} \to \mathcal{E}$ such that $F\gamma = \delta$, and since δ is also universal, by (1) there is a unique functor $G : \mathcal{E} \to \operatorname{Ho} \mathcal{C}$ such that $G\delta = \gamma$. Left composing by G on our first relation gives

$$F\gamma = \delta \implies GF\gamma = G\delta.$$

Since $G\delta = \gamma$, we have that $GF\gamma = \gamma$. Similarly, left composing our second relation by F gives $FG\delta = \delta$. Finally, note that the identity $1_{\operatorname{Ho} \mathcal{C}} : \operatorname{Ho} \mathcal{C} \to \operatorname{Ho} \mathcal{C}$ is the unique arrow induced by universality of γ with respect to itself[a]; in other words $1_{\operatorname{Ho} \mathcal{C}}$ is the unique arrow such that $1_{\operatorname{Ho} \mathcal{C}}\gamma = \gamma$. Since we also have that $GF\gamma = \gamma$, we conclude that $GF = 1_{\operatorname{Ho} \mathcal{C}}$. The exact same proof for the other relation shows that $FG = 1_{\mathcal{E}}$, and thus F and G are isomorphisms.

Finally, define the isomorphism by $F \mapsto \operatorname{Ho} F$. This is obviously well-defined, but since the definition of $\operatorname{Ho} F$ is uniquely forced by F, it has a well-defined inverse $\operatorname{Ho} F \mapsto F$ given simply by precomposition by γ. Now let $\tau : F \Rightarrow G$; we must define $\operatorname{Ho} \tau : \operatorname{Ho} F \Rightarrow \operatorname{Ho} G$. We may freely define it componentwise as $\operatorname{Ho} \tau_x = \tau_x$, since the target categories are the same in both functor categories. We need only show $\operatorname{Ho} \tau_x$ is natural with respect to inverses weak equivalences, since τ is natural on F and G, and $\operatorname{Ho} F, \operatorname{Ho} G$ agree with F, G outside of weak equivalences. This is easiest seen algebraicly:

$$\tau_y F(w) = G(w)\tau_x \implies \tau_y = G(w)\tau_x F(w)^{-1} \implies G(w)^{-1}\tau_y = \tau_x F(w)^{-1} \implies \operatorname{Ho} G(w)^{-1}\tau_y = \tau_x \operatorname{Ho} F(w)^{-1}.$$

Vertical composition of natural transformations is preserved manifestly, and horizontal composition is a nonfactor since this is a functor category. ■

[a] Since γ inverts weak equivalences, by universality of γ there is an arrow $\operatorname{Ho} \mathcal{C} \to \operatorname{Ho} \mathcal{C}$ as defined in part (1).

Remark 1.2.5. After working through the algebra for (3) and recalling a proof I did in enriched categories, I am beginning to think that if one has an isomorphism, commutivity of any diagram is preserved by swapping an arrow with its isomorphism. Although I don't see how to prove this in the general case, and as such it may not hold in the general case, I can see an obvious proof if the isomorphism is on the "end" of the diagram, meaning there are either no arrows postcomposing with it, or no arrows precomposing with it. One then simply notes that if they wrote out the algebraic expressions the diagram represents, the isomorphism is always either on the front or back of the equation, and one can simply compose on the relevant side with the inverse.

Proposition 1.2.6. *Let $\mathcal{C}$ be a model category, and write $\mathcal{C}_c$, $\mathcal{C}_f$, and $\mathcal{C}_{cf}$ for the full subcategories of cofibrant, fibrant, and cofibrant and fibrant objects of $\mathcal{C}$. Then the inclusion functors induce equivalences of categories $\operatorname{Ho} \mathcal{C}_{cf} \to \operatorname{Ho} \mathcal{C}_c \to \operatorname{Ho} \mathcal{C}$ and $\operatorname{Ho} \mathcal{C}_{cf} \to \operatorname{Ho} \mathcal{C}_f \to \operatorname{Ho} \mathcal{C}$.*

Proof. The referenced functors are induced as follows: the localizations in the diagram

$$\begin{tikzcd} \mathcal{C}_{cf} \arrow[r] \arrow[d] & \mathcal{C}_c \arrow[r] \arrow[d] & \mathcal{C} \arrow[d] \\ \operatorname{Ho}\mathcal{C}_{cf} & \operatorname{Ho}\mathcal{C}_c & \operatorname{Ho}\mathcal{C} \end{tikzcd}$$

invert weak equivalences, and thus the composition of a localization with the inclusion inverts weak equivalences. Universality as in lemma 1.2.4 induces functors

$$\begin{tikzcd} \mathcal{C}_{cf} \arrow[r] \arrow[d] & \mathcal{C}_c \arrow[r] \arrow[d] & \mathcal{C} \arrow[d] \\ \operatorname{Ho}\mathcal{C}_{cf} \arrow[r, dashed] & \operatorname{Ho}\mathcal{C}_c \arrow[r, dashed] & \operatorname{Ho}\mathcal{C} \end{tikzcd}$$

as required. We first show the inverse of the inclusion $i : \mathcal{C}_c \hookrightarrow \mathcal{C}$ is given by cofibrant replacement. We have $(Q \circ i)(x) = Q(x)$, and since $Q(x)$ is cofibrant, this is a valid functor to $\mathcal{C}_c$. Since x is also guarenteed to be cofibrant since $x \in \mathcal{C}_c$, and since $\mathcal{C}_c$ is full, the component $q_x = \beta(0 \to x) : Q(x) \to x$ as defined in remark 1.1.16 exists in $\mathcal{C}_c$ for each cofibrant x. Since $Qi(x) = Q(x)$, by remark 1.1.16, this gives a natural transformation $q : Qi \to 1_{\mathcal{C}_c}$. Moreover, this transformation is pointwise trivial fibrant. The functors lower to

$$\operatorname{Ho}\mathcal{C}_c \underset{\operatorname{Ho} Q}{\overset{\operatorname{Ho} i}{\rightleftarrows}} \operatorname{Ho}\mathcal{C}$$

and q lowers to $\operatorname{Ho} q : \operatorname{Ho} Q \circ \operatorname{Ho} i \Rightarrow 1_{\operatorname{Ho}\mathcal{C}_c}$ by lemma 1.2.4. Since $\operatorname{Ho} q_x = q_x$ and q_x is a trivial fibration, and in particular is a weak equivalence in $\mathcal{C}_c$, we have that q_x is an isomorphism in $\operatorname{Ho}\mathcal{C}_c$ since localization inverts weak equivalences. Thus, since each component of q_x is an isomorphism, we have a natural isomorphism $\operatorname{Ho} q : \operatorname{Ho} Q \circ \operatorname{Ho} i \cong 1_{\operatorname{Ho}\mathcal{C}_c} = \operatorname{Ho} 1_{\mathcal{C}_c}$. The converse direction is the same, since $iQ = Q$, so it follows directly by this argument and remark 1.1.16.

Now we show the inclusion $i : \mathcal{C}_{cf} \hookrightarrow \mathcal{C}_c$ induces an equivalence of categories. The inverse will in this case be given by fibrant replacement and cofibrant replacement. Note that each $x \in \mathcal{C}_{cf}$ is cofibrant and fibrant by assumption. We have that $(R \circ i)(x) = R(x)$ is fibrant, but we must also show it is cofibrant. Note that x is cofibrant, so the arrow $0 \to x$ is a cofibration. By the axioms of a model category, the arrow $\gamma(x \to *) : x \to R(x)$ is a trivial cofibration, and in particular a cofibration. Composing these gives an arrow $0 \to R(x)$ which is a cofibration, and thus $R(x)$ is cofibrant. Now note that there is a natural trivial cofibration $r : 1_{\mathcal{C}} \Rightarrow R$ by remark 1.1.16. Since $r_x : x \to R(x)$ has fibrant domain and codomain, it exists in $\mathcal{C}_{cf}$, and thus we obtain a natural trivial cofibration $r : 1_{\mathcal{C}_{cf}} \Rightarrow Ri$. By the same logic as above, localization inverts r into a natural isomorphism. The converse direction is once again the same, since once again $iR = R$. ■

Remark 1.2.7. An interesting result from my argument is that $R(x)$ preserves cofibrancy, and dually $Q(x)$ preserves fibrancy. In other words, if x is fibrant, then so is $Q(x)$, and if x is cofibrant, then so is $R(x)$.

We thus have that $\operatorname{Ho}\mathcal{C}_{cf} \simeq \operatorname{Ho}\mathcal{C}$. There is an alternative construction of $\operatorname{Ho}\mathcal{C}_{cf}$ for model categories for which hom-sets are indeed sets, which we now summarize. We denote by $\Delta : x \to x \times x$ the diagonal map and $\nabla : x \amalg x \to x$ the dual map, called the *fold map*.

Definition 1.2.8 (Homotopy)**.** Let $\mathcal{C}$ be a model category, and let $f, g : b \to x$ be maps in $\mathcal{C}$.

1. A *cylinder object* for b is a factorization of the fold map $\nabla : b \amalg b \to b$ into a cofibration $b \amalg b \xrightarrow{i_0+i_1} b'$ followed by a weak equivalence $b' \xrightarrow{s} b$;
2. A *path object* for x is a factorization of the diagonal map $\Delta : x \to x \times x$ into a weak equivalence $x \xrightarrow{r} x'$ followed by a fibration $x' \xrightarrow{(p_0,p_1)} x \times x$;
3. A *left homotopy* from f to g is a map $H : b' \to x$ for some cylinder object b' for b such that $Hi_0 = f$ and $Hi_1 = g$. We say that f and g are *left homotopic*, written $f \sim^{\ell} g$, if there is a left homotopy from f to g;
4. A *right homotopy* from f to g is a map $K : b \to x'$ for x' some path object for x, such that $p_0K = f$ and $p_1K = g$. We say that f and g are *right homotopic*, written $f \sim^{r} g$, if there is a right homotopy from f to g;
5. We say that f and g are *homotopic*, written $f \sim g$, if they are both left and right homotopic;
6. f is a *homotopy equivalence* when there is some $h : x \to b$ such that $hf \sim 1_b$ and $fh \sim 1_x$.

♡

Intuition 1.2.9. The reason we refer to cylinder objects as cylinders is most naturally seen after developing intuition for a left homotopy. Diagammatically, a left homotopy H from f to g is a commutative diagram

$$\begin{tikzcd}
b \arrow[d] \arrow[dr, "i_0"] \arrow[drr, bend left, "f"] & & \\
b \amalg b \arrow[r, "i_0+i_1"] & b' \arrow[r, "H"] & x \\
b \arrow[u] \arrow[ur, "i_1"'] \arrow[urr, bend right, "g"'] & &
\end{tikzcd}$$

where $i_0 + i_1$ is a cofibration, and b' is weakly equivalent to b as noted above. Ignore for a moment the cofibration and weak equivalence semantics. Imagine b' as an actual cylinder $B \times I$ for B a topological space and I the unit interval. If $f, g : B \to X$ are continuous maps, a regular homotopy between them is a continuous map $H : B \times I \to X$ which agrees with F on $B \times \{0\} \cong B$ and agrees with g on $B \times \{1\} \cong B$. Although these subspaces are isomorphic, they are disjoint as subsets of the cylinder $B \times I$. Thus the intuition is that b in the diagram is the domain space of the morphisms, and b' is the cylinder $b \times I$. The inclusion i_0 inserts b into the bottom face of the cylinder at 0, and i_1 includes it at the top face at 1. A homotopy is intuitively thought of as a map along the entire cylinder to x, agreeing with f and g on the given faces, and this is indeed what the diagram above expresses. Of course there is not necessarily a unit interval object I in $\mathcal{C}$, so this is a further generalization of this notion. The weak equivalence puts a constraint on how far this generalization goes. In topology, the cylinder is homotopy equivalent to any of its faces by contracting along I; in particular it is homotopy equivalent to b. The weak equivalence $b' \to b$ enforces this rule in model categories as well. Finally, I am unsure why $i_0 + i_1$ need be a cofibration, but my intuition is starting to be that cofibrations are an axiomatic generalization of monomorphisms, in which case this makes total sense as well. We will have to see going forward if this intuition holds.

Once again, we begin by considering the diagrammatic representation of a right homotopy, before discussing the term *path object*. A right homotopy from f to g is a diagram

$$\begin{tikzcd}
& & x \\
b \arrow[urr, bend left, "f"] \arrow[r, "H"] \arrow[drr, bend right, "g"'] & x' \arrow[ur, "p_0"] \arrow[r, "(p_0,p_1)"] \arrow[dr, "p_1"'] & x \times x \arrow[u] \arrow[d] \\
& & x
\end{tikzcd}$$

where (p_0, p_1) is a fibration, and x' is weakly equivalent to x as noted above. Note the obvious fact that path objects are dual to cylinder objects, and right homotopies are dual to left homotopies. Once again, we imagine B, X as topological spaces. We will see that this notion of homotopy is less familiar from topology, and more familiar from simplicial homotopy in a cotensored simplicial category. The path object in this case should be thought of as the path space X^I. The maps p_0 and p_1 are then thought of as evaluating paths in X^I at 0, and evaluating paths in X^I at 1. The map (p_0, p_1) is then just $\gamma \mapsto (\gamma(0), \gamma(1))$. A homotopy is then a continuous map $B \to X^I$, such that evaluating at 0 agrees with f and evaluating at 1 agrees with g. It can be thought of as encoding the information of f and g into the path space by evaluating along the unit interval. The weak equivalence once again guarentees the homotopy equivalence $X^I \sim X$, which follows since evaluation is continuous, and $X^* \cong X$ for any singleton $* \subseteq I$. The fact that (p_0, p_1) is a fibration implies that one may lift homotopies along the map from $X \times X$ to X^I.

Remark 1.2.10. By duality, left homotopies in $\mathcal{C}$ are right homotopies in $D\mathcal{C}$ and vice versa, and the same with path and cylinder objects. Thus, one need only prove statements about left homotopies and cylinder objects in order to prove them for path objects and right homotopies.

Remark 1.2.11. One immediately considers the applications of the functorial factorizations to the construction of path objects and cylinder objects. Indeed, apply the functorial factorization (α, β) to $\nabla : b \amalg b \to b$ to obtain a cylinder object $Q(b \amalg b)$, with a cofibration $b \amalg b \to Q(b \amalg b)$ and a trivial fibration $Q(b \amalg b) \to b$. This factorization has the added bonus of the weak equivalence being a fibration, and the cylinder object being cofibrant. The book denotes this object by $b \times I$, which is purely notation and does not indicate an actual product. In some enriched categorical settings, this can however be literal instead of notation, such as topological or simplicial categories. Dually, we obtain a path object $R(x)$ for x by the other factorization, which has the benefit of being fibrant, and the weak equivalence being a cofibration. We will sometimes denote this object as x^I when viewing it as a path object.

Proposition 1.2.12. *Let $\mathcal{C}$ be a model category, and $f, g : b \to x$ maps in $\mathcal{C}$.*

1. *If $f \sim^\ell g$ and $h : x \to y$, then $hf \sim^\ell hg$. Dually, if $f \sim^r g$ and $h : a \to b$, then $fh \sim^r gh$;*
2. *If x is fibrant, $f \sim^\ell g$, and $h : a \to b$, then $fh \sim^\ell gh$. Dually, if b is cofibrant, $f \sim^r g$, and $h : x \to y$, then $hf \sim^r hg$;*
3. *If b is cofibrant, then left homotopy is an equivalence relation on $\mathcal{C}(b, x)$. Dually, if x is fibrant, then right homotopy is an equivalence relation on $\mathcal{C}(b, x)$;*
4. *If b is cofibrant and $h : x \to y$ is a trivial fibraiton or a weak equivalence of fibrant objects, then h induces an isomorphism*
$$\mathcal{C}(b, x)/{\sim^\ell} \cong \mathcal{C}(b, y)/{\sim^\ell} .$$
Dually, if x is fibrant and $h : a \to b$ is a trivial cofibration or a weak equivalence of cofibrant

objects, then h induces an isomorphism

$$\mathcal{C}(b,x)/{\sim^r} \cong \mathcal{C}(a,x)/{\sim^r};$$

5. *If b is cofibrant, then $f \sim^\ell g$ implies $f \sim^r g$. Furthermore, if x' is any path object for x, then there is a right homotopy $K : b \to x'$ from f to g. Dually, if x is fibrant, then $f \sim^r g$ implies $f \sim^\ell g$, and there is a left homotopy from f to g using any cylinder object for b.*

Proof. By remark 1.2.10, we need only prove the statements for left homotopies and cylinder objects. The statement of (1) is fairly obvious, given a homotopy $f \sim^\ell g$, consider the diagram

$$\begin{tikzcd}
b \arrow[d] \arrow[dr, "i_0"] \arrow[drr, bend left, "f"] & & & \\
b \amalg b \arrow[r, "i_0+i_1"] & b' \arrow[r, "H"] & x \arrow[r, "h"] & y \\
b \arrow[u] \arrow[ur, "i_1"'] \arrow[urr, bend right, "g"'] & & &
\end{tikzcd}$$

One immediately sees that composing along h gives hH a homotopy from hf to hg.

To prove (2), let

$$b \amalg b \xrightarrow{i} b' \xrightarrow{s} b$$

be the cylinder object. Note that s is a weak equivalence, and s admits functorial factorizations into a cofibration and a trivial fibration, or a trivial cofibration and a fibration. Note that for either choice, we have that one of the components of the factorization is a weak equivalence, and since the composition is a weak equivalence, by the 2-out-of-3 property we have that the third map is a weak equivalence. Thus s factors as a trivial cofibration followed by a trivial fibration

$$b' \xrightarrow{k} b'' \xrightarrow{s'} b.$$

Since $k(i_0+i_1)$ is a composition of cofibrations, we may consider b'' to be a cylinder object for b. Because x is fibrant, the homotopy $H : b' \to x$ extends to a homotopy $H : b'' \to x$. We may in particular use the factorization (γ, δ) to choose b'' to be fibrant by remark 1.1.16, giving a diagram

$$\begin{tikzcd}
b' \arrow[r, tail, "k"] \arrow[rr, bend left, "s"] \arrow[d, "H"'] & b'' \arrow[r, two heads, "s'"] \arrow[ddl, two heads] & b \\
x \arrow[d, two heads] & & \\
* & &
\end{tikzcd}$$

where I have denoted fibrations by $\twoheadrightarrow$ and cofibrations by $\rightarrowtail$. Note that the triangle commutes by universality of $*$, and we may rewrite this commutative triangle as the commutative square of solid arrows

$$\begin{tikzcd}
b' \arrow[r, "H"] \arrow[d, tail, "k"'] & x \arrow[d, two heads] \\
b'' \arrow[r, two heads] \arrow[ur, dotted, "H'"] & 1
\end{tikzcd}$$

Since k is a trivial cofibration and $x \to *$ a fibration, and since trivial cofibrations have the left lifting property with respect to all fibrations, the dotted arrow as indicated exists, rendering the diagram commutative. Since $k(i_0 + i_1)$ and b'' form a cylinder object as noted above, and $H'k = H$, we have that the homotopy H does indeed extend to a homotopy H' for f and g. With this understanding, we may without loss of generality take the arrow from the cylinder object to b to be a trivial fibration.

Quickly note that $h\nabla_a = \nabla_b(h \amalg h)$, which is an identity that holds in generality. We see this by comparing the two diagrams

$$\begin{tikzcd}
a \arrow[d] \arrow[dr, "1"] & & \\
a \amalg a \arrow[r, "\nabla_a"] & a \arrow[r, "h"] & b \\
a \arrow[u] \arrow[ur, "1"'] & &
\end{tikzcd}
\qquad
\begin{tikzcd}
a \arrow[r, "h"] \arrow[d] & b \arrow[d] \arrow[dr, "1"] & \\
a \amalg a \arrow[r, "h \amalg h"] & b \amalg b \arrow[r, "\nabla_b"] & b \\
a \arrow[u] \arrow[r, "h"'] & b \arrow[u] \arrow[ur, "1"'] &
\end{tikzcd}$$

and noting that composing along the h gives the same arrows $a \to b$ in both diagrams, so by universality the compositions $a \amalg a \to b$ must be the same. Noting this fact, now let

$$a \amalg a \xrightarrow{j} a' \xrightarrow{t} a$$

be a cylinder object for a, which exists by remark 1.2.11. We note that the diagram

$$\begin{tikzcd}
a \amalg a \arrow[r, "h \amalg h"] \arrow[d, "j"'] \arrow[dr, dotted, "\nabla_a"] & b \amalg b \arrow[r, "i_0 + i_1"] \arrow[dr, dotted, "\nabla_b"] & b' \arrow[d, "s"] \\
a' \arrow[r, "t"'] & a \arrow[r, "h"'] & b
\end{tikzcd}$$

commutes, since the definition of a cylinder object guarentees $s(i_0 + i_1) = \nabla_b$ and $tj = \nabla_a$, hence the dotted lines, and the middle square expresses the equality $\nabla_b(h \amalg h) = h\nabla_a$. Thus we obtain that the diagram of solid lines

$$\begin{tikzcd}
a \amalg a \arrow[r, "(i_0+i_1)(h \amalg h)"] \arrow[d, "j"'] & b' \arrow[d, "s"] \\
a' \arrow[r, "ht"'] \arrow[ur, dotted, "u"] & b
\end{tikzcd}$$

commutes. Since j is cofibrant by definition of a cylinder object, and s is fibrant by our earlier assumption, we obtain a lift, indicated by the dotted line, rendering the diagram commutative.

Let H be the homotopy from f to g. We claim that Hu is a homotopy fh to fg. Indeed, since

$$\begin{tikzcd}
a \arrow[r, "h"] \arrow[d] & b \arrow[d] \arrow[dr, "i_0"] \arrow[drr, bend left, "f"] & & \\
a \amalg a \arrow[r, "h \amalg h"] & b \amalg b \arrow[r, "i_0+i_1"] & b' \arrow[r, "H"] & x \\
a \arrow[u] \arrow[r, "h"'] & b \arrow[u] \arrow[ur, "i_1"'] \arrow[urr, bend right, "g"'] & &
\end{tikzcd}$$

commutes, and since $uj = (i_0 + i_1)(h \amalg h)$ by the lifting diagram, we replace these arrows and obtain that the diagram

$$\begin{tikzcd}
a \arrow[r, "h"] \arrow[d] \arrow[dr] & b \arrow[dr, "i_0"] \arrow[drr, bend left, "f"] & & \\
a \amalg a \arrow[r, "j"] & a' \arrow[r, "u"] & b' \arrow[r, "H"] & x \\
a \arrow[u] \arrow[ur] \arrow[r, "h"'] & b \arrow[ur, "i_1"'] \arrow[urr, bend right, "g"'] & &
\end{tikzcd}$$

commutes, where the arrows $a \to a'$ arise by universality, and thus uH is indeed a left homotopy $fh \sim^{\ell} gh$, as desired.

Now consider (3). Note that $\sim^{\ell}$ is obviously symmetric, since we need only flip the diagram horizontally. Showing $\sim^{\ell}$ is reflexive is also simple: by proposition 1.2.12, there is a cylinder object $\nabla = s \circ (i_0 + i_1)$ given by a functorial factorization, and postcomposing by f gives a diagram

$$\begin{tikzcd}
b \arrow[d] \arrow[dr, "i_0"] \arrow[drrr, bend left, "f"] & & & \\
b \amalg b \arrow[r, "i_0+i_1"] & b' \arrow[r, "s"] & b \arrow[r, "f"] & x \\
b \arrow[u] \arrow[ur, "i_1"'] \arrow[urrr, bend right, "f"'] & & &
\end{tikzcd}$$

and thus fs is a homotopy $f \sim^{\ell} f$. To show it's transitive, we must now make use of the hypothesis that b be cofibrant. Let $H : b' \to x$ be a homotopy from f to g, and let $H' : b'' \to x$ be a homotopy from g to h. Let $i_0 + i_1$ be the relevant map corresponding to H, and $i_0' + i_1'$ the map corresponding to H'. Define c via the pushout square

$$\begin{tikzcd}
b \arrow[r, "i_1"] \arrow[d, "i_0'"'] & b' \arrow[d] \\
b'' \arrow[r] & c
\end{tikzcd}$$

and define j_0, j_1 as the compositions

$$
\begin{aligned}
j_0 &: b \xrightarrow{i_0} b' \longrightarrow c, \\
j_1 &: b \xrightarrow{i_1'} b'' \longrightarrow c.
\end{aligned}
$$

We want $j_0 + j_1 : b \amalg b \to c$ to define a cylinder object, but we need the first map to be a cofibration, and we need a weak equivalence $t : c \to b$. We will focus on the second one for now. Define t as the map induced by universality of pushout in the diagram

$$
\begin{tikzcd}
b \arrow[r, "i_1"] \arrow[d, "i_0'"'] & b' \arrow[d] \arrow[ddr, bend left, "s"] & \\
b'' \arrow[r] \arrow[drr, bend right, "s'"'] & c \arrow[dr, dashed, "t"] & \\
& & b
\end{tikzcd}
$$

and note that b is cofibrant. Note that if we can show i_1 is a trivial cofibration, then by corollary 1.1.20 so is $b'' \to c$, and since $t \circ (b'' \to c) = s'$ and s' is a weak equivalence, we have that by the 2-of-3 property t is a weak equivalence.

Note that by universality of 0, the pushout of $b \leftarrow 0 \to b$ is the coproduct

$$
\begin{tikzcd}
0 \arrow[r] \arrow[d] & b \arrow[d] \\
b \arrow[r] & b \amalg b
\end{tikzcd}
$$

and by corollary 1.1.20 the inclusions into the coproduct are cofibrations. Since $i_1 = (i_0 + i_1) \circ \iota_1$ for $\iota_1 : b \to b \amalg b$ the relevant inclusion is a composition of cofibrations, we have that i_1 is a cofibration. By remark 1.1.19, all isomorphisms are trivial cofibrations; in particular the identity. Since $s \circ i_1 = 1_b$, and since s and 1_b are weak equivalences, by 2-out-of-3 so is i_1.

Note here that

$$
b \amalg b \xrightarrow{j_0 + j_1} c \xrightarrow{t} b
$$

is a factorization of the fold map, since j_0 factors along si_0:

$$
\begin{tikzcd}
b \arrow[d] \arrow[dr, "i_0"] \arrow[drr, bend left, "j_0"] & & & \\
b \amalg b \arrow[r, "i_0 + i_1"] & b' \arrow[r] \arrow[rr, bend right, "s"'] & c \arrow[r, "t"] & b \\
b \arrow[u] \arrow[ur, "i_1"'] & & &
\end{tikzcd}
$$

commutes by inspection of the pullback diagram and the defintions of j_0 and t. The same can be said for j_1. Thus we obtain $tj_0 = si_0 = 1_b$, and the same for j_1. We would have our cylinder now, exceopt for the fact that $j_0 + j_1$ need not be a cofibration. Before we fix this, note that the homotopies H and H' induce a map $K : c \to x$ along the pullback:

$$\begin{tikzcd}
b \arrow[r, "i_1"] \arrow[d, "i'_0"'] & b' \arrow[d] \arrow[ddr, bend left, "H"] & \\
b'' \arrow[r] \arrow[drr, bend right, "H'"'] & c \arrow[dr, dashed, "K"] & \\
& & x
\end{tikzcd}$$

Note that once again by inspection of the pullback diagram and by the definitions of j_0 and j_1, the diagrams

$$\begin{tikzcd}
f : b \arrow[r, "i_0"] \arrow[rr, bend right, "j_0"'] & b' \arrow[r] \arrow[rr, bend left, "H"] & c \arrow[r, "K"'] & x
\end{tikzcd}$$

$$\begin{tikzcd}
h : b \arrow[r, "i'_1"] \arrow[rr, bend right, "j_1"'] & b'' \arrow[r] \arrow[rr, bend left, "H"] & c \arrow[r, "K"'] & x
\end{tikzcd}$$

commute, and thus we have $Kj_0 = f$ and $Kj_1 = h$. One would hope that we can turn K into a homotopy by finding a cofibration to replace $j_0 + j_1$, and indeed we can. Factor $j_0 + j_1$ into a cofibration and a trivial fibration $b \amalg b \to c' \to c$. We can show that c' is a cylinder object for b, and that the composition

$$c' \longrightarrow c \xrightarrow{K} x$$

is a left homotopy between f and h. Indeed, since the arrow $c' \to c$ is a *trivial* fibration, it is in particular a weak equivalence, and thus composing with t gives another weak equivalence. Thus we have a weak equivalence $c' \to b$. Denote by z the cofibration $b \amalg b \to c'$. We then have a diagram

$$\begin{tikzcd}
b \arrow[d, "\iota_0"'] \arrow[dr, "z\iota_0"] \arrow[drr, bend left, "j_0"] & & & \\
b \amalg b \arrow[r, "z"] & c' \arrow[r] & c \arrow[r, "K"] & x \\
b \arrow[u, "\iota_1"] \arrow[ur, "z\iota_1"'] \arrow[urr, bend right, "j_1"'] & & &
\end{tikzcd}$$

which commutes, where I have labeled the inclusions simply for clarity. Since all we did was factor $j_0 + j_1$, we still have that the composition is the fold map, and thus we do indeed have the stated homotopy, as required.

Next, we show (4). We start with the trivial fibration case. Let $h : x \to y$ be a trivial fibration, and define the map

$$F : \mathcal{C}(b, x)/\sim^{\ell} \to \mathcal{C}(b, y)/\sim^{\ell}$$

by postcomposition $F[f] = [hf]$, which is well-defined by (1) and (3). We first show F is surjective. Let $f' : b \to y$ be some map. Since b is cofibrant and h is a trivial fibration, the lifting problem indicated by the diagram of solid arrows

$$\begin{tikzcd}
0 \arrow[r] \arrow[d] & x \arrow[d, "h"] \\
b \arrow[r, "f'"'] \arrow[ur, dotted, "f"] & y
\end{tikzcd}$$

admits a dotted arrow rendering the diagram commutative. We thus have that there exists $f : b \to x$ such that $hf = f'$, and the operation is surjective without even passing to homotopy classes. Now let $H : b' \to y$ be a left homotopy $hf \sim^{\ell} hg$. Thus we have a diagram

$$\begin{tikzcd}
b \arrow[r, "f"] \arrow[d] \arrow[dr, "i_0"] & x \arrow[dr, "h"] & \\
b \amalg b \arrow[r, "i_0+i_1"] & b' \arrow[r, "H"] & y \\
b \arrow[r, "g"'] \arrow[u] \arrow[ur, "i_1"'] & x \arrow[ur, "h"'] &
\end{tikzcd}$$

and one notes that $hf+hg = h(f+g)$ by universality, and $hf+hg = H(i_0+i_1)$ also by universality. One thus constructs the diagram of solid arrows

$$\begin{tikzcd}
b \amalg b \arrow[r, "f+g"] \arrow[d, "i_0+i_1"'] & x \arrow[d, "h"] \\
b' \arrow[r, "H"'] \arrow[ur, dotted, "K"] & y
\end{tikzcd}$$

and since h is a trivial fibration and $i_0 + i_1$ is a cofibration, there is a lift indicated by the dotted arrow rendering the diagram commutative. Note that the lift K is indeed homotopy from f to g by universality of coproduct, and thus F is injective. Therefore, F is indeed an isomorphism.

Now we must consider the case where $h : x \to y$ is a weak equivalence between fibrant objects. One notes that we might hope to view h as a natural transformation $h : \mathcal{C}(-, x)/{\sim^{\ell}} \Rightarrow \mathcal{C}(-, y)/{\sim^{\ell}}$. By assumption, the domain and codomain are fibrant objects, and thus they exist in $\mathcal{C}_f$. By (1) and (3), the quotient operation is compatible with composition. Thus we have a functor

$$x \xmapsto{y} \mathcal{C}_{cf}(-, x) \longmapsto \mathcal{C}_{cf}(-, x)/{\sim^{\ell}}$$

given by taking the Yoneda embedding of $\mathcal{C}_f$, restricing the resultant functors to $\mathcal{C}_{cf}$ to ensure the domain is cofibrant as well, then composing with the quotient map. The notation $\mathcal{C}_{cf}(-, x)$ is thus an abuse of notation, since x need not be cofibrant; we simply write this to indicate the arguments the functor can take must be fibrant and cofibrant. By the first case, if $h : x \to y$ is a trivial fibration, then if we can show it is natural, there will be an induced isomorphism $\mathcal{C}_{cf}(-, x) \cong \mathcal{C}_{cf}(-, y)$. By taking weak equivalences in the target category to be isomorphisms, we have that trivial fibrations, in particular trivial fibrations between fibrant objects, are taken to weak equivalences. By Ken Brown's lemma 1.1.21, this expands to all weak equivalences, proving our statement. It thus suffices to show the given definition of h is natural. Indeed, we have that for any $f : b \to c$,

$$\begin{tikzcd}
\mathcal{C}(c,x)/\sim^{\ell} \arrow[r, "h_*"] \arrow[d, "f^*"'] & \mathcal{C}(c,y)/\sim^{\ell} \arrow[d, "f^*"] \\
\mathcal{C}(b,x)\sim^{\ell} \arrow[r, "h_*"'] & \mathcal{C}(b,y)\sim^{\ell}
\end{tikzcd}$$

commutes simply by definition of precomposition and postcomposition. Thus, the statement is proven.

Finally, we show (5). Suppose b is cofibrant and $f \sim^{\ell} g$ by a homotopy $H : b' \to x$. We must show $f \sim^{r} g$ using any path object. Since b is cofibrant, the maps i_0, i_1 are trivial cofibrations by the same argument used in part (3) with the pushout of $0 \to b$. Let

$$x \xrightarrow{r} x' \xrightarrow{(p_0,p_1)} x \times x$$

be a path object for x. Note that since $Hi_0 = f$ and $si_0 = 1_b$, the diagram

$$\begin{tikzcd}
 & & b \arrow[dd, "f"] \\
b \arrow[r, "i_0"] \arrow[urr, equal, bend left] & b' \arrow[ur, "s"] \arrow[dr, "H"'] & \\
 & & x
\end{tikzcd}$$

commutes. By universality of product, one concludes that $(fs, H)i_0 = (fsi_0, Hi_0) = (f, f)$ by this diagram. Similarly, note that since $p_0 r = p_1 r = 1_x$, we have that

$$\begin{tikzcd}
 & & x \\
b \arrow[r, "f"] & x \arrow[r, "r"] \arrow[ur, equal] \arrow[dr, equal] & x' \arrow[u, "p_0"'] \arrow[d, "p_1"] \\
 & & x
\end{tikzcd}$$

commutes, and by universality once again we have that $(p_0, p_1)rf = (f, f)$. We conclude that the diagram of solid arrows

$$\begin{tikzcd}
b \arrow[r, "rf"] \arrow[d, "i_0"'] & x' \arrow[d, "(p_0,p_1)"] \\
b' \arrow[r, "(fs,H)"'] \arrow[ur, dotted, "J"] & x \times x
\end{tikzcd}$$

commutes. Since i_0 is trivial cofibrant and (p_0, p_1) is fibrant, there exists a lift as indicated by the dotted arrow rendering the diagram commutative. Finally, we show that $K = Ji_1$ is a right homotopy from f to g. Indeed, by the above diagram

$$(p_0, p_1)Ji_1 = (p_0, p_1)(fs, H)i_1 = (p_0 fsi_1, p_1 Hi_1) = (p_0 f, p_1 g)$$

since $Hi_1 = g$ and $si_1 = 1$. Thus the restriction along p_0 is f and the restriction along p_1 is g, and Ji_1 is a homotopy, as required. ∎

Corollary 1.2.13. *Let $\mathcal{C}$ be a model category. If b is cofibrant and x is fibrant, then the left and right homotopy relations coencide, and are equivalence relations on $\mathcal{C}(b, x)$. Moreover, if $f \sim g : b \to x$, then there is a left homotopy $H : b' \to x$ from f to g using any cylinder object b' for b. Dually, there is a right homotopy $k : b \to x'$ from f to h using any path object x' for b.*

Proof. Indeed, that the relations coencide follows by proposition 1.2.12 (5), the fact that they are equivalence relations follows from (4), and the invariance under cylinder/path objects is (5) once again. ∎

Corollary 1.2.14. *The homotopy relation on the morphisms of $\mathcal{C}_{cf}$ is an equivalence relation and is compatible with composition. Hence the category $\mathcal{C}_{cf}/\!\sim$ exists.*

Remark 1.2.15. From our proof to proposition 1.2.12 (3), we saw that if b is cofibrant, than the inclusions $b \to b \amalg b$ are cofibrations, and thus the arrows $i_0, i_1 : b \to b'$ are trivial cofibrations. Dually, the arrows $p_0, p_1 : x' \to x$ are trivial fibrations when x is fibrant.

Recall our result from proposition 1.2.6 that $\operatorname{Ho}\mathcal{C}_{cf} \simeq \operatorname{Ho}\mathcal{C}$. It would definitely be nice to show that the quotient category by the homotopy relation is equivalent to $\operatorname{Ho}\mathcal{C}_{cf}$, and thus to $\operatorname{Ho}\mathcal{C}$, since this branch of math is literally called *homotopy theory*. Our first step in proving this would be to show that $\mathcal{C}_{cf}/\!\sim$ inverts weak equivalences, thus at least getting us a functor $\operatorname{Ho}\mathcal{C}_{cf} \to \mathcal{C}_{cf}/\!\sim$ by universality of localization. We accomplish this in the following proposition.

Proposition 1.2.16. *Let $\mathcal{C}$ be a model category. Then a map in $\mathcal{C}_{cf}$ is a weak equivalence if and only if it is a homotopy equivalence. In particular, quotienting by homotopy inverts weak equivalences.*

Proof. First let $f : a \to b$ be a weak equivalence of cofibrant and fibrant objects. By proposition 1.2.12, we have a natural isomorphism $f_* : (\mathcal{C}_{cf}/\!\sim)(x, a) \to (\mathcal{C}_{cf}/\!\sim)(x, b)$ for any fibrant and cofibrant x. Choose $x = b$; there then necessarily exists a map $g : b \to a$ which maps by postcomposition by f to 1_b; hence $fg \sim 1_b$. By proposition 1.2.12, we then have $fgf \sim f$, so taking $x = a$ now gives $fgf \sim f$ in the codomain, and thus there is some k such that postcomposition $fk \sim fgf \sim f$. Necessarily by injectivity, k admits a definition as both gf and 1, and thus $gf \sim 1$. Thus, f is a homotopy equivalence.

The converse is more involved. Let f be a homotopy equivalence between cofibrant and fibrant objects. Factor f into a trivial cofibration $g : a \to c$ and a fibration $p : c \to b$. Clearly c is cofibrant, since the unique arrow from $0 \to c$ factors as $g \circ (0 \to a)$, which is a composition of cofibrations. Similarly, $(b \to *) \circ p$ is a composition of fibrations, so c is cofibrant and fibrant. By the proof of the other direction, since g is a trivial cofibration and in particular a weak equivalence, it is a homotopy equivalence. By the 2-out-of-3 property, it suffices to show p is a weak equivalence. Let $f' : b \to a$ be a homotopy inverse to f, and let $H : b' \to b$ be the left homotopy $ff' \sim 1_b$. Since

$f = pg$, one factors $ff' = pgf' = Hi_0$, and constructs the commutative diagram of solid arrows:

$$\begin{tikzcd}
b \arrow[r, "gf'"] \arrow[d, "i_0"'] & c \arrow[d, "p"] \\
b' \arrow[r, "H"'] \arrow[ur, dotted, "H'"] & b
\end{tikzcd}$$

Since b is cofibrant, by remark 1.2.15 i_0 is a trivial cofibration, and since p is a fibration, there is a dotted arrow as indicated rendering the diagram commutative. Define $q = H'i_1$, and note that $pH' = H$ implies that $pq = 1_b$, since H is a homotopy $ff' \sim 1_b$. Note also that since $H'i_0 = gf'$, we have that H' is a homotopy from gf' to q. Since we found earlier that g was a homotopy equivalence, let g' be a homotopy inverse for g. Then $p \sim pgg' \sim fg'$. We thus conclude that

$$qp \sim (gf')(fg') \sim 1_c.$$

Let $K : c' \to c$ be a left homotopy from 1_c to qp, and note that $Ki_0 = 1_c$ is a weak equivalence since it is an isomorphism. By remark 1.2.15, i_0 is a weak equivalence, and by the 2-out-of-3 property, so is K. Since i_1 is a weak equivalence by the same remark, we thus have that $Ki_1 = qp$ is also a weak equivalence. Recalling that $pq = 1_b$, we have that the diagram

$$\begin{tikzcd}
c \arrow[r, equal] \arrow[d, "p"'] & c \arrow[r, equal] \arrow[d, "qp"'] & c \arrow[d, "p"] \\
b \arrow[r, "q"'] & c \arrow[r, "p"'] & b
\end{tikzcd}$$

commutes. We also have that the top and bottom rows are the identity, and thus p is a retract of qp. By the retract axiom, p is a weak equivalence. ∎

Corollary 1.2.17. *Let $\mathcal{C}$ be a model category, and let $\gamma : \mathcal{C}_{cf} \to \operatorname{Ho}\mathcal{C}_{cf}$ and $\delta : \mathcal{C}_{cf} \to \mathcal{C}_{cf}/\sim$ be the canonical functors. Then there is a unique isomorphism $j : \mathcal{C}_{cf}/\sim\, \cong \operatorname{Ho}\mathcal{C}_{cf}$ such that $j\delta = \gamma$. In particular, j is the identity on objects.*

Proof. By universality, it suffices to show $\mathcal{C}_{cf}/\sim$ has the same universal property as described in lemma 1.2.4. We know already that δ inverts weak equivalences by proposition 1.2.16. Let $F : \mathcal{C}_{cf} \to \mathcal{D}$ be a functor which inverts weak equivalences. Let

$$a \amalg a \xrightarrow{i_0 + i_1} a' \xrightarrow{s} a$$

be a cylinder object for a. Then $si_0 = si_1 = 1_a$ by assumption. Since s is a weak equivalence, and F inverts weak equivalences,

$$F(1_a) = F(s)F(i_0) = F(s)F(i_1) \implies F(s)^{-1}F(s)F(i_0) = F(i_0) = F(i_1) = F(s)^{-1}F(s)F(i_1).$$

Thus, if $H : a' \to b$ is a left homotopy between $f, g : a \to b$, then

$$F(f) = F(H)F(i_0) = F(H)F(i_1) = F(g).$$

Thus, F identifies homotopic maps. By universality of quotient categories, there is a unique functor $\mathcal{C}_{cf}/\!\sim \to \mathcal{D}$ as required.

Note that we did not show a' was cofibrant and fibrant. We need not show this for $a \amalg a$, since we did not compute a functor on an arrow with $a \amalg a$ as domain or codomain. We define a' by a functorial factorization, and by proposition 1.2.12 it does not matter which cylinder object we choose, so this is fine. By remark 1.2.15 the arrow $i_0 : a \to a'$ is a cofibration, and since $0 \to a \to a'$ is a composition of cofibrations, a' is cofibrant. Since a is fibrant, and $a' \to a$ is a fibration, we have that a' is fibrant as well, concluding the proof. ■

Theorem 1.2.18. *Let $\mathcal{C}$ be a model category. Let $\gamma : \mathcal{C} \to \operatorname{Ho}\mathcal{C}$ be the localization functor, and let Q and R denote the cofibrant and fibrant replacements of $\mathcal{C}$, respectively, as defined in remark 1.1.16.*

1. *The inclusion $\mathcal{C}_{cf} \to \mathcal{C}$ induces an equivalence of categories*

$$\mathcal{C}_{cf}/\!\sim\; \cong \operatorname{Ho}\mathcal{C}_{cf} \cong \operatorname{Ho}\mathcal{C};$$

2. *There are natural isomorphisms*

$$\mathcal{C}(QR(x), QR(y))/\!\sim\; \cong \operatorname{Ho}\mathcal{C}(\gamma(x), \gamma(y)) \cong \mathcal{C}(RQ(x), RQ(y))/\sim.$$

   In addition, there is a natural isomorphism

$$\operatorname{Ho}\mathcal{C}(\gamma(x), \gamma(y)) \cong \mathcal{C}(Q(x), R(y))/\!\sim,$$

   and if x is cofibrant and y is fibrant, there is a natural isomorphism

$$\operatorname{Ho}\mathcal{C}(\gamma(x), \gamma(y)) \cong \mathcal{C}(x, y).$$

   In particular, $\operatorname{Ho}\mathcal{C}$ is a category without passing to a higher universe;

3. *The functor $\gamma : \mathcal{C} \to \operatorname{Ho}\mathcal{C}$ identifies left and right homotopic maps;*
4. *If $f : a \to b$ is a map in $\mathcal{C}$ such that $\gamma(f)$ is an isomorphism in $\operatorname{Ho}\mathcal{C}$, then f is a weak equivalence.*

Proof. (1) follows by corollary 1.2.17 and proposition 1.2.6. Investigation of the inverses as noted in proposition 1.2.6 gives the natural isomorphisms of (2). The rest of (2) follows from the natural weak equivalences $Q \Rightarrow 1 \Rightarrow R$ as noted in remark 1.1.16, and invoking proposition 1.2.12 (3,5). Part (3) is subsumed by corollary 1.2.17. For (4), let $f : a \to b$ be a map in $\mathcal{C}$ which is inverted by γ. Then $QR(f)$ is an isomorphism in $\mathcal{C}_{cf}/\!\sim$, and thus $QR(f)$ is necessarily a homotopy equivalence. By corollary 1.2.17, $QR(f)$ is a weak equivalence. Since $Q \Rightarrow 1$ is a natural weak equivalence,

$$\begin{tikzcd} QR(a) \arrow[r, "q_{R(a)}"] \arrow[d, "QR(f)"'] & R(a) \arrow[d, "R(f)"] \\ QR(b) \arrow[r, "q_{R(b)}"'] & R(b) \end{tikzcd}$$

commutes, and in particular $QR(f)$ and $q_{R(b)}$ are weak equivalences. Thus $q_{R(b)}QR(f) = R(f)q_{R(a)}$ is a weak equivalence, and since $q_{R(a)}$ is also a weak equivalence, so is $R(f)$. The same argument with $1 \Rightarrow R$ shows f is a weak equivalence. ■

Remark 1.2.19. We now have a variety of remarkable facts. Localization inverts precisely those maps that are weak equivalences; Riehl calls categories for which this is true *saturated*. We also have that weak equivalences are precisely homotopy equivalences, which is a remarkable fact and implies that model categories provide excellent structure for homotopy theory in a general setting. We have shown that homotopy categories of model categories are locally small, and that we can even choose to work in the category given by quotienting by homotopy, if we are willing to wrangle the inverse maps when necessary. We also have that localization identitifies homotopic maps, another excellent cue that model categories are an excellent setting for homotopy theory. Finally, the book notes it will denote $\operatorname{Ho}\mathcal{C}(x, y)$ by $[x, y]$ as we continue through the text. I will likely adopt this convention as well since $\operatorname{Ho}\mathcal{C}(x, y)$ is quite annoying to type, but I don't like the square brackets much either, and really think we should come up with a new notation.

Chapter 2

Weak factorization systems in model categories (Riehl 11)

We now turn to a more modern exposition on model categories from Emily Riehl's *Categorical Homotopy Theory*. The presentation will be more terse, since the theory is already well documented in Hovey, but it provides a new viewpoint based on weak factorization systems.

2.1 Lifting problems and lifting properties

A *lifting problem* between arrows i and f in $\mathcal{M}$ is a commutative diagram of solid arrows

$$\begin{tikzcd} \cdot \arrow[r, "u"] \arrow[d, "i"'] & \cdot \arrow[d, "f"] \\ \cdot \arrow[r, "v"'] \arrow[ur, dotted] & \cdot \end{tikzcd}$$

A *lift* or *solution* is a dotted arrow as indicated, rendering the diagram commutative. If all lifting problems between i and f admit solutions, we say i has the left lifting property with respect to f, and f has the right lifting property with respect to i. I will write $i \boxtimes f$ to denote this. Riehl manages to get a box with a slash in it, which I have no idea how to typeset since `\boxslash` is not a command. If $\mathcal{L}$ is a class of maps in $\mathcal{M}$, we write $\mathcal{L}^{\boxtimes}$ for the class of arrows that have the *right lifting property* with respect to all arrows in $\mathcal{L}$, and similarly we write ${}^{\boxtimes}\mathcal{R}$ for the set of all arrows that have the left lifting property with respect to all morphisms in $\mathcal{R}$.

For example, consider the diagram of solid arrows

$$\begin{tikzcd} \varnothing \arrow[r] \arrow[d] & x \arrow[d, "f"] \\ * \arrow[r] \arrow[ur, dotted] & y \end{tikzcd}$$

in $\mathcal{S}\mathrm{et}$. This commutes manifestly by universality of $\varnothing$. A lift would also make the left triangle commute manifestly, so consider only the right triangle. The arrow $* \to y$ selects an element of y, and the composition $* \to x \to y$ selects an element in the image of f. Thus, for the diagram to commute, the element selected by $* \to y$ must be in the image of f. In particular, for f to have the right lifting property against $\varnothing \to *$, it must be surjective, and thus epi.

We can also consider the diagram of solid arrows

$$\begin{tikzcd} * \amalg * \arrow[r] \arrow[d] & x \arrow[d, "f"] \\ * \arrow[r] \arrow[ur, dotted] & y \end{tikzcd}$$

for some $f : x \to y$. Note that the map $* \amalg * \to x$ selects two points in x, and $* \to y$ selects one point in y. For the diagram to commute, these points must be identified with each other. A lift $* \to x$ selects an element of x, so for the diagram to commute with a lift, any two points selected by $* \amalg * \to *$ must be equal to the element $* \to x$. Thus, f has the right lifting property wrt $* \amalg * \to *$ precisely when it only identifies points if they are equal, hence f is injective and mono.

Example 2.1.1. A more interesting example is the Hurewicz (co)fibrations. Writing I for the unit interval, write i_0 for the inclusion $A \to A \times I$ at $0 \in I$, and write p_0 for the projection $A^I \to A$ at $0 \in I$. We define the *Hurewicz fibrations* as the set of maps $\{i_0 : A \to A \times I \mid A \in \mathcal{T}\mathrm{op}\}^{\boxtimes}$ with the right lifting property thereof, and similarly we define the Hurewicz cofibrations as ${}^{\boxtimes}\{p_0 : A^I \to A \mid A \in \mathcal{T}\mathrm{op}\}$. Note these are not in general sets, but they do indeed form classes, which is sufficient. Restricting only to disks gives the class of *Serre fibrations* $\{i_0 : D^n \to D^n \times I \mid n \in \mathbb{N}\}^{\boxtimes}$.

Lemma 2.1.2. *Any class of arrows of the form ${}^{\boxtimes}\mathcal{R}$ is closed under coproducts, pushouts, transfinite composition, retracts, and contains the isomorphisms. We call such a class* weakly saturated.

Proof. Let i have the left lifting property with respect to $\mathcal{R}$, and let j be an arrow admitting a retract

$$\begin{tikzcd}
\cdot \arrow[r, "s_1"] \arrow[d, "j"'] \arrow[rr, bend left, "1"] & \cdot \arrow[r, "r_1"] \arrow[d, "i"] & \cdot \arrow[d, "j"] \\
\cdot \arrow[r, "s_2"'] \arrow[rr, bend right, "1"'] & \cdot \arrow[r, "r_2"'] & \cdot
\end{tikzcd}$$

and let $r \in \mathcal{R}$. One must show that j admits a lift against all arrows in $\mathcal{R}$, so let

$$\begin{tikzcd}
\cdot \arrow[r, "u"] \arrow[d, "j"'] & \cdot \arrow[d, "r"] \\
\cdot \arrow[r, "v"'] & \cdot
\end{tikzcd}$$

be a lifting problem with $r \in \mathcal{R}$. Gluing this to the retract gives the diagram of solid arrows

$$\begin{tikzcd}
\cdot \arrow[r, "s_1"] \arrow[d, "j"'] \arrow[rr, bend left, "1"] & \cdot \arrow[r, "r_1"] \arrow[d, "i"'] & \cdot \arrow[r, "u"] \arrow[d, "j"'] & \cdot \arrow[d, "r"] \\
\cdot \arrow[r, "s_2"'] \arrow[rr, bend right, "1"'] & \cdot \arrow[r, "r_2"'] \arrow[urr, dotted, "h"] & \cdot \arrow[r, "v"'] & \cdot
\end{tikzcd}$$

Since i has the left lifting property with respect to r, there is a dotted arrow as indicated rendering the diagram commutative. Note that one has by commutivity that

$$hs_2j = ur_1s_1 = u,$$

and thus we might hope hs_2 is a solution to the original lifting problem. We need only show $rhs_2 = v$, and indeed, by commutivity we have

$$rhs_2 = vr_2s_2 = v.$$

Thus, j has the left lifting property with respect to r, and ${}^{\boxslash}\mathcal{R}$ is closed under retracts. The others have already been proven, or I don't feel like proving them. ■

Remark 2.1.3. The dual statement of lemma 2.1.2 is that maps in $\mathcal{L}^{\boxslash}$ are closed under products, pullbacks, transfinite composition, retracts, and contain th e isomorphisms.

Notation 2.1.4. If $\mathcal{L}, \mathcal{R}$ are classes of maps, we write $\mathcal{L} \boxslash \mathcal{R}$ to denote that $\mathcal{L} \subseteq^{\boxslash} \mathcal{R}$ and $\mathcal{R} \subseteq \mathcal{L}^{\boxslash}$. Obvserve that $(-)^{\boxslash}$ and ${}^{\boxslash}(-)$ form a Galois connection with respect to inclusion, which is a case of an adjunction and is kinda like adjunctions with subset-like functors.

Lemma 2.1.5. *Let $F : \mathcal{M} \rightleftarrows \mathcal{N} : U$ be an adjunction and let $\mathcal{A}$ be a class of arrows in $\mathcal{M}$, and $\mathcal{B}$ a class of arrows in $\mathcal{N}$. Then $F(\mathcal{A}) \boxslash \mathcal{B}$ if and only if $\mathcal{A} \boxslash U(\mathcal{B})$.*

Proof. Adjunctions $F \dashv U$ raise to adjunctions $F : \operatorname{Map} \mathcal{M} \rightleftarrows \mathcal{N} : U$ between the arrow categories. Lifting problems are identified by transposing under this adjunction, and naturality guarentees that transposing a solution gives a solution. ■

Construction 2.1.6 (Leibniz). Consider a two-variable adjunction, which is a collection of functors

$$- \otimes - : \mathcal{M} \times \mathcal{N} \to \mathcal{P}, \qquad - \pitchfork - : \mathcal{M}^{\mathrm{op}} \times \mathcal{P} \to \mathcal{N}, \qquad \underline{\hom}(-,-) : \mathcal{N}^{\mathrm{op}} \times \mathcal{P} \to \mathcal{M}$$

satisfying the adjunction relations

$$\mathcal{P}(m \otimes n, p) \cong \mathcal{N}(n, m \pitchfork p) \cong \mathcal{M}(m, \underline{\hom}(n,p)).$$

Suppose that $\mathcal{P}$ has pushouts and $\mathcal{M}, \mathcal{N}$ have pullbacks. The Leibniz construction then provides us a two-variable adjunction

$$-\widehat{\otimes}- : \mathcal{M}^2 \times \mathcal{N}^2 \to \mathcal{P}^2, \qquad -\widehat{\pitchfork}- : \left(\mathcal{M}^2\right)^{\mathrm{op}} \times \mathcal{P}^2 \to \mathcal{N}^2, \qquad \widehat{\underline{\hom}}(-,-) : \left(\mathcal{N}^2\right)^{\mathrm{op}} \times \mathcal{P}^2 \to \mathcal{M}^2$$

between the arrow categories $(-)^2 \cong \operatorname{Map}(-) \cong (1_{(-)} \downarrow 1_{(-)})$. They are defined as follows: let $i : m \to m'$ be an arrow in $\mathcal{M}^2$, and let $j : n \to n'$ an arrow in $\mathcal{N}^2$. The *Leibniz tensor*, also called the *pushout-product*, is defined by the unique dashed arrow in the pushout diagram

$$\begin{tikzcd}
m \otimes n \arrow[r, "i\otimes 1"] \arrow[d, "1\otimes j"'] & m' \otimes n \arrow[d] \arrow[ddr, bend left, "1\otimes j"] & \\
m \otimes n' \arrow[r] \arrow[drr, bend right, "i\otimes 1"'] & \cdot \arrow[dr, dashed, "i\widehat{\otimes} j"] & \\
& & m' \otimes n'
\end{tikzcd}$$

Similarly, the Leibniz cotensor and Leibniz hom, also known as the pullback-cotensor and pullback-hom, are given by pullbacks in $\mathcal{M}$ and $\mathcal{N}$ Let $f : p \to p'$ in $\mathcal{P}$:

$$\begin{tikzcd}
m' \pitchfork p \arrow[dr, dashed, "i \widehat{\pitchfork} f"] \arrow[drr, bend left, "i \pitchfork 1"] \arrow[ddr, bend right, "1 \pitchfork f"'] & & \\
& \cdot \arrow[r] \arrow[d] & m \pitchfork p \arrow[d, "1 \pitchfork f"] \\
& m' \pitchfork p' \arrow[r, "i \pitchfork 1"'] & m \pitchfork p'
\end{tikzcd}
\qquad
\begin{tikzcd}
\underline{\hom}(n', p) \arrow[dr, dashed, "\widehat{\underline{\hom}}(j,f)"] \arrow[drr, bend left, "\underline{\hom}(j,1)"] \arrow[ddr, bend right, "\underline{\hom}(1,f)"'] & & \\
& \cdot \arrow[r] \arrow[d] & \underline{\hom}(n, p) \arrow[d, "\underline{\hom}(1,f)"] \\
& \underline{\hom}(n', p') \arrow[r, "\underline{\hom}(j,1)"'] & \underline{\hom}(n, p')
\end{tikzcd}$$

Lemma 2.1.7. *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be classes of maps in $\mathcal{M}, \mathcal{N}, \mathcal{P}$, respectively. Then*

$$\mathcal{A} \widehat{\otimes} \mathcal{B} \boxtimes \mathcal{C} \iff \mathcal{B} \boxtimes \mathcal{A} \widehat{\pitchfork} \mathcal{C} \iff \mathcal{A} \boxtimes \widehat{\underline{\hom}}(\mathcal{B}, \mathcal{C}).$$

Proof. Once again, lifting problems transpose into other lifting problems, and so do solutions. ■

2.2 Weak factorization systems

Definition 2.2.1 (Weak Factorization System). A *weak factorization system* on a category is a pair $(\mathcal{L}, \mathcal{R})$ of morphisms such that

1. (factorization) every arrow admits a factorization as an arrow of $\mathcal{L}$ followed by an arrow of $\mathcal{R}$;
2. (lifting) $\mathcal{L} \boxtimes \mathcal{R}$;
3. (closure) $\mathcal{L} = {}^{\boxtimes}\mathcal{R}$ and $\mathcal{R} = \mathcal{L}^{\boxtimes}$.

♡

Remark 2.2.2. The third axiom obviously subsumes the second, and it means that any class in a weak factorization system determines the other. Weak factorization systems lay the groujndwork for a more modern view of model category theory, with a greater focus on the factorizations maps admit into cofibrations and fibrations.

Lemma 2.2.3. *In the presence of the first two axioms, one can replace the closure axiom by (closure'):* the classes $\mathcal{L}$ and $\mathcal{R}$ are closed under retracts.

Proof. lemma 2.1.2 proves that closure implies closure'. For the converse, note that any $k \in {}^{\boxtimes}\mathcal{R}$ factors as $r\ell$ for $r \in \mathcal{R}$ and $\ell \in \mathcal{L}$. By definition, k has the left lifting property with respect to r, and by lemma 1.1.17 k is a retract of ℓ. Since $\mathcal{L}$ is closed under retracts, $k \in \mathcal{L}$. Thus, $\mathcal{L} \subseteq {}^{\boxtimes}\mathcal{R}$, and the converse inclusion follows by the second axiom (lifting). ■

Remark 2.2.4. Oftentimes one has that a weak factorization system is generated from a set $\mathcal{J}$ by taking the right class to be $\mathcal{J}^{\boxtimes}$ and the left class to be ${}^{\boxtimes}(\mathcal{J}^{\boxtimes})$. Indeed $({}^{\boxtimes}(\mathcal{J}^{\boxtimes}))^{\boxtimes} = \mathcal{J}^{\boxtimes}$. We will see that if the category is subject to the small object argument, then the factorizations produced are appropriate for a model category. We call the category *cofibrantly generated* in that case.

2.3 Model categories and Quillen functors

Riehl presents the following alternative to definition 1.1.5, and we spend a large portion of this section dedicated to proving the equivalence of these definitions. The book continues by giving a lot of examples, but since I will cover some these examples much more indepth in Hovey after reading about Quillen functors, I will pass on them for now.

Definition 2.3.1 (Model Structure)**.** A *model structure* on a complete and cocomplete homotopical category $(\mathcal{M}, \mathcal{W})$ is two classes of morphisms $\mathcal{C}, \mathcal{F}$ such that $(\mathcal{C} \cap \mathcal{W}, \mathcal{F})$ and $(\mathcal{C}, \mathcal{W} \cap \mathcal{F})$ are weak factorization systems. Maps in $\mathcal{C}$ are called *cofibrations* and maps in $\mathcal{F}$ are called *fibrations*. ♡

Remark 2.3.2. Riehl requires homotopical categories to satisfy the 2-out-of-6 axiom, which is stronger than the 2-out-of-3 axiom, and says for any composable triple (f, g, h), if hg and gf are weak equivalences, then all of f, g, h, hgf also are. By theorem 1.2.18 (4), model categories are saturated, meaning the maps inverted by localization are precisely the weak equivalences. Riehl shows that *any* collection of arrows $\mathcal{W}$ for which the localization is saturated necessarily satisfies the 2-out-of-6 property. The proof is fairly simple; since the category is saturated, isomorphisms in $\operatorname{Ho}\mathcal{M}$ are precisely the weak equivalences, and since isomorphisms are easily shown to satisfy the 2-out-of-6 property, the weak equivalences necessarily do as well.

Remark 2.3.3. By lemma 2.2.3, cofibrations and fibrations are closed under retracts. Since the 2-out-of-6 axiom is stronger, the 2-out-of-3 axiom is satisfied. Definition 2.2.1 gives that every cofibration has the left lifting property against trivial fibrations, and every trivial cofibration has the left lifting property against fibrations. We can also show that weak factorization systems are functorial factorizations: let $(\mathcal{L}, \mathcal{R})$ be a weak factorization system for some category $\mathcal{C}$, and let $(u, v) : f \to g$ in $\operatorname{Map}\mathcal{C}$. Factoring $f = r_f \ell_f$ and $g = r_g \ell_g$ gives a diagram

$$\begin{tikzcd}
x \arrow[r, "u"] \arrow[d, "\ell_f"'] & a \arrow[d, "\ell_g"] \\
e_f \arrow[d, "r_f"'] & e_g \arrow[d, "r_g"] \\
y \arrow[r, "v"'] & b
\end{tikzcd}$$

which rearranges to the diagram of solid arrows

$$\begin{tikzcd}
x \arrow[r, "u"] \arrow[d, "\ell_f"'] & a \arrow[r, "\ell_g"] & e_g \arrow[d, "r_g"] \\
e_f \arrow[r, "r_f"'] \arrow[ur, dotted] & y \arrow[r, "v"'] & b
\end{tikzcd}$$

Since $\ell_f \in \mathcal{L}$ has the left lifting property with respect to all elements of $\mathcal{R}$, and since $r_g \in \mathcal{R}$, the dotted arrow as indicated exists, rendering the diagram commutative. Since the factorization also held u and v fixed, one can indeed view this factorization as a functorial factorization in the sense of definition 1.1.1. Clearly, a full model category also admits weak factorization systems, as described, via its functorial factorizations. It remains only to be shown that $\mathcal{W}$ is closed under retracts.

Lemma 2.3.4. *Let* $(\mathcal{C} \cap \mathcal{W}, \mathcal{F})$ *and* $(\mathcal{C}, \mathcal{F} \cap \mathcal{W})$ *be weak factorization systems on a complete and* *cocomplete category $\mathcal{M}$, and let $\mathcal{W} \subseteq \mathcal{M}$ satisfy the 2-out-of-3 property. Then $\mathcal{W}$ is closed under retracts.*

Proof. Let $f : x \to y$ be a retract of some weak equivalence $w : a \to b$. Factor f as hg with g a trivial cofibration. The retract diagram then becomes

$$\begin{tikzcd}
x \arrow[r, "h_1"] \arrow[d, "g"'] \arrow[rr, bend left, "1"] & a \arrow[r, "k_1"] \arrow[dd, "w"'] & x \arrow[d, "g"] \\
e_f \arrow[d, "h"'] & & e_f \arrow[d, "h"] \\
y \arrow[r, "h_2"'] \arrow[rr, bend right, "1"'] & b \arrow[r, "k_2"'] & y
\end{tikzcd}$$

Pushout along the left corner of the diagram $e_f \xleftarrow{g} x \xrightarrow{h_1} a$ to obtain the diagram of solid arrows

$$\begin{tikzcd}
x \arrow[r, "h_1"] \arrow[d, "g"'] \arrow[rr, bend left, "1"] & a \arrow[r, "k_1"] \arrow[d, "i"'] \arrow[dd, bend left, "w"] & x \arrow[d, "g"] \\
e_f \arrow[r] \arrow[d, "h"'] & e_f \times_x a \arrow[d, dashed, "j"'] & e_f \arrow[d, "h"] \\
y \arrow[r, "h_2"'] \arrow[rr, bend right, "1"'] & b \arrow[r, "k_2"'] & y
\end{tikzcd}$$

Note that since w has domain a and sh has domain e_f, and since they both have the same codomain, they form a cone for the pushout diagram. Hence by universality there is a unique dashed arrow as indicated, rendering the diagram commutative. Note by lemma 2.1.2 that since g is a trivial cofibration, so is i. In particular, $i \in \mathcal{W}$. Since $w \in \mathcal{W}$ as well, and $ji = w$, by the 2-of-3 property j is a weak equivalence. Note that we also have gr and 1_{e_f} form a cone for the pushout as well, so the dashed arrow ℓ in the below diagram

$$\begin{tikzcd}
x \arrow[r, "h_1"] \arrow[d, "g"'] \arrow[rr, bend left, "1"] & a \arrow[r, "k_1"] \arrow[d, "i"'] & x \arrow[d, "g"] \\
e_f \arrow[r, "m"'] \arrow[d, "h"'] & e_f \times_x a \arrow[r, dashed, "\ell"'] \arrow[d, dashed, "j"'] & e_f \arrow[d, "h"] \\
y \arrow[r, "h_2"'] \arrow[rr, bend right, "1"'] & b \arrow[r, "k_2"'] & y
\end{tikzcd}$$

exists, and by universality the diagram commutes (one could also construct a cone to b in the bottom right corner, and then universality guarentees commutivity much more obviously, but this is unnecessary). One notes also by commutivity of the relevant pushout diagram, in particular in the triangle formed by the identity, that we obtain $\ell m = 1$; thus the lower two rows show that h is a retract of j. If we can show h is a weak equivalence, then since g is also a weak qeuivalence, $f = hg$ is also a weak equivalence. Thus it suffices to show h is a weak equivalence. Recall that h is a fibration, and that we have shown j is a weak equivalence. It thus suffices to show if a retract of a weak equivalence is a fibration, then it is a weak equivalence.

Let f be a fibration, and let f be a retract of some weak equivalence w. Factor $w = vu$ by any weak factorization, and note that by 2-of-3, both v and u are weak equivalences. We thus have the diagram of solid arrows

$$\begin{tikzcd}
x \arrow[r, "uh"] \arrow[dd, "f"'] \arrow[rr, bend left, "1"] & a \arrow[r, "h"] \arrow[d, "u"'] & x \arrow[dd, "f"] \\
& \cdot \arrow[ur, dotted, "t"'] & \\
y \arrow[r] \arrow[rr, bend right, "1"'] & b \arrow[r, "k"'] & y
\end{tikzcd}$$

Note that by viewing kv as the bottom of a lifting problem, we have that since u is a trivial cofibration and f is a fibration, the dotted arrow t as indicated exists, rendering the diagram commutative. By commutativity, $ts = 1$, and thus f is a retract of v. Since v is a trivial cofibration, by remark 2.3.2 so is f. Thus the statement is proven. ■

Example 2.3.5. Although I have chosen to pass on the examples given here in order to pursue them more indepth in Hovey, I think the following example is interesting enough to warrant writing down. The category $\mathcal{C}at$ admits a model structure, where weak equivalences are categorical equivalences, cofibrations are functors which are injective on objects, and fibrations are *isofibrations*, which are functors that have the right lifting property with respect to any inclusion into the *free standing isomorphism*. This is an object in $\mathcal{C}at$ defined as the groupoid $\mathcal{I} = \{0 \leftrightarrows 1\}$. A functor $\mathcal{I} \to \mathcal{C}$ is thus just a choice of an isomorphism in $\mathcal{C}$. Investigating what this implies, consider an isofibration

$F : \mathcal{C} \to \mathcal{D}$, and a lifting problem

$$\begin{tikzcd} * \arrow[r, "c_0"] \arrow[d, "0"'] & \mathcal{C} \arrow[d, "F"] \\ \mathcal{I} \arrow[r, "d_0 \cong d_1"'] \arrow[ur, dotted] & \mathcal{D} \end{tikzcd}$$

where the notation of the diagram should make clear what is happening, but I will explain it anyways. We without loss of generality take the inclusion $* \hookrightarrow \mathcal{I}$ to select the object 0, and we note that the arrow $* \to \mathcal{C}$ selects an element c_0 of $\mathcal{C}$. For a lift to exist, we simply require there be a map $I \to \mathcal{C}$ such that $0 \mapsto c_0$; indeed since $c_0 \cong c_0$ we always have such a map. Thus the left triangle tells us nothing interesting. The functor $\mathcal{I} \to \mathcal{D}$ maps $(0, 1) \mapsto (d_0, d_1)$, as should be obvious by the notation. Thus the lifting problem gives us that $F(c_0) = d_0$. For a lift to exist, we must have an arrow $\ell : \mathcal{I} \to \mathcal{C}$ as indicated by the dotted line; equivalently an isomorphism $c_0 \cong c_1$ in $\mathcal{C}$, such that $F(c_1) = d_1$. Thus a lift exists whenever every d_1 isomorphic to $d_0 \in F(\mathcal{C})$ has at least one preimage under F which is isomorphic to at least one preimage of d_0 under F. Thus an isofibration is simply a functor F for which every isomorphism $d \cong F(c)$ for some $c \in \mathcal{C}$ lifts to an isomorphism $c' \cong c$ in $\mathcal{C}$.

Chapter 3

Quillen functors and derived functors (Hovey 1.3)

We now introduce Quillen functors, the natural notion of morphisms between model categories. One would assume that a morphism of model categories should preserve cofibrations, fibrations, and weak equivlances, but in practice this is generally too strict of a requirement. The reasoning for which is more easily described after we have seen definition 3.1.1 and have proven lemma 3.1.4, so explanation of this reasoning has been delegated to remark 3.1.6.

We will additionally see that Quillen functors associate an *derived functor* to the homotopy category, and the process of associating Quillen functors to derived functors is functorial up to natural isomorphism; it is thus a weak 2-functor. When this derived functor is an equivalence of categories, we call the corresponding Quillen functors *Quillen equivalences*, for reasons which are already fairly clear, but will be detailed in [link].

3.1 Quillen functors

Definition 3.1.1 (Quillen Functor)**.** Let $\mathcal{C}$ and $\mathcal{D}$ be model categories.

1. A functor $F : \mathcal{C} \to \mathcal{D}$ is a *left Quillen functor* if F is left adjoint, and preserves cofibrations and trivial cofibrations;
2. Dually, a functor $U : \mathcal{D} \to \mathcal{C}$ is a *right Quillen functor* if U is right adjoint, and preserves fibrations and trivial fibrations;
3. An adjunction $F : \mathcal{C} \rightleftarrows \mathcal{D} : U$ is a *Quillen adjunction* if F is a left Quillen functor.

♡

Remark 3.1.2. By Ken Brown's lemma 1.1.21, left Quillen functors preserve weak equivalences between cofibrant objects, and right Quillen functors preserve weak equivalences between fibrant objects. Hovey notes that most of the results in this section require only that left Quillen functors preserve cofibrant objects and weak equivalences between them, but he also notes that not much is gained by this generality, implying not many functors in practice that satisfy such a property are not Quillen. Note also that the definition of a Quillen adjunction requires only that F be left Quillen; this is reconsiled in lemma 3.1.4.

Notation 3.1.3. A Quillen adjunction $F : \mathcal{C} \rightleftarrows : U$ is often referred to only by mentioning one of the adjoint functors. We write φ for the adjunct map, $\eta : 1_{\mathcal{C}} \Rightarrow UF$ for the unit, and $\varepsilon : FU \Rightarrow 1_{\mathcal{D}}$ for the counit of the adjunction. We prefer to refer to adjunctions either by (F, U, φ) or by $(F, U, \eta, \varepsilon)$, especially when the domains and codomains of F and U can be understood. We will often say (F, U, φ) *is an adjunction from* $\mathcal{C}$ *to* $\mathcal{D}$ and will write $(F, U, \varphi) : \mathcal{C} \to \mathcal{D}$ to indicate that $F : \mathcal{C} \to \mathcal{D}$ and $U : \mathcal{D} \to \mathcal{C}$. We will also, of course, write $F \dashv U$ when specifying any of $\varphi, \eta, \varepsilon$ is not necessary.

Lemma 3.1.4. *Let $(F, U, \varphi) : \mathcal{C} \to \mathcal{D}$ be an adjunction between model categories. Then (F, U, φ) is Quillen if and only if U is right Quillen. In particular, F is left Quillen if and only if U is right Quillen.*

Proof. First, we show $F(f)$ has the left lifting property with respect to p if and only if f has the left lifting property with respect to $U(p)$. We then specialize to cofibrations and fibrations, and apply lemma 1.1.18.

Let $f : c \to c'$ in $\mathcal{C}$ and let $p : d \to d'$ in $\mathcal{D}$. Let $F(f)$ have the left lifting property with respect to p, and let

$$\begin{tikzcd} c \arrow[r, "h"] \arrow[d, "f"'] & U(d) \arrow[d, "U(p)"] \\ c' \arrow[r, "k"'] & U(d') \end{tikzcd}$$

be a lifting problem. By functoriality of F, the left square

$$\begin{tikzcd} F(c) \arrow[r, "F(h)"] \arrow[d, "F(f)"'] & FU(d) \arrow[r, "\varepsilon_d"] \arrow[d, "FU(p)"] & d \arrow[d, "p"] \\ F(c') \arrow[r, "F(k)"'] \arrow[urr, dotted] & FU(d') \arrow[r, "\varepsilon_{d'}"'] & d' \end{tikzcd}$$

commutes, and by naturality of $\varepsilon : FU \Rightarrow 1_{\mathcal{D}}$, the diagram on the right commutes. Denote the dotted arrow by ℓ. Since $F(f)$ has the left lifting property with respect to p, there exists a dotted arrow as indicated rendering the diagram commutative. By functoriality of U, the right two squares in the diagram

$$\begin{tikzcd} c \arrow[r, "\eta_c"] \arrow[d, "f"'] & UF(c) \arrow[r, "UF(h)"] \arrow[d, "UF(f)"'] & UFU(d) \arrow[r, "U(\varepsilon_d)"] \arrow[d, "UFU(p)"'] & U(d) \arrow[d, "p"] \\ c' \arrow[r, "\eta_{c'}"'] & UF(c') \arrow[r, "UF(k)"'] \arrow[urr, dotted] & UFU(d') \arrow[r, "U(\varepsilon_{d'})"'] & U(d') \end{tikzcd}$$

commute, and by naturality of $\eta : 1_{\mathcal{C}} \Rightarrow UF$, the square on the left commutes. It would be helpful to show that the composites along the top and bottom rows simplified to h and k. Consider the square

$$\begin{tikzcd} c \arrow[r, "\eta_c"] \arrow[d, "h"'] & UF(c) \arrow[d, "UF(h)"] \\ U(d) \arrow[r, "\eta_{U(d)}"'] & UFU(d) \end{tikzcd}$$

which commutes by naturality of η. Recall also that $U\varepsilon \circ \eta U = 1$, and thus $U(\varepsilon_d) \circ \eta_{U(d)} = 1$.

Thus, the diagram

$$\begin{tikzcd}
& U(d) \arrow[dr, "\eta_{U(d)}"] \arrow[drr, bend left, "1"] & & \\
c \arrow[ur, "h"] \arrow[r, "\eta_c"] \arrow[d, "f"'] & UF(c) \arrow[r, "UF(h)"] & UFU(d) \arrow[r, "U(\varepsilon_d)"] & U(d) \arrow[d, "p"] \\
c' \arrow[r, "\eta_{c'}"'] & UF(c') \arrow[r, "UF(k)"'] \arrow[urr, dotted, "U(\ell)"] & UFU(d') \arrow[r, "U(\varepsilon_{d'})"'] & U(d')
\end{tikzcd}$$

commutes as well, and the top row simplifies to h. Note that we have dropped the middle arrows for simplicity. By a similar argument, the bottom row simplifies to k: by naturality of η, the square

$$\begin{tikzcd}
c' \arrow[r, "\eta_{c'}"] \arrow[d, "k"'] & UF(c') \arrow[d, "UF(k)"] \\
U(d') \arrow[r, "\eta_{U(d')}"'] & UFU(d')
\end{tikzcd}$$

commutes, and thus the diagram

$$\begin{tikzcd}
c \arrow[rrr, "h"] \arrow[d, "f"'] & & & U(d) \arrow[d, "p"] \\
c' \arrow[r, "\eta_{c'}"'] \arrow[urrr, dotted, "U(\ell)\circ\eta_{c'}"] \arrow[dr, "k"'] & UF(c') \arrow[r, "UF(k)"'] \arrow[urr, dotted, "U(\ell)"'] & UFU(d') \arrow[r, "U(\varepsilon_{d'})"'] & U(d') \\
& U(d') \arrow[ur, "\eta_{U(d')}"'] \arrow[urr, bend right, "1"'] & &
\end{tikzcd}$$

also commutes. Note that I have also composed $U(\ell) \circ \eta_{c'}$ to obtain a lift over the main square, so simplifying the diagram gives that the square

$$\begin{tikzcd}
c \arrow[r, "h"] \arrow[d, "f"'] & U(d) \arrow[d, "U(p)"] \\
c' \arrow[r, "k"'] \arrow[ur, dotted, "U(\ell)\circ\eta_{c'}"] & U(d')
\end{tikzcd}$$

commutes. Thus, f has the left lifting property with respect to $U(p)$. The converse proof is extremely similar.

Now we conclude the proof with repeated references to lemma 1.1.18.

1. ($\implies$)

   (a) Let F be left Quillen. Then F preserves all cofibrations. Let p be a trivial fibration and let f be a cofibration. Since F preserves cofibrations, we have that $F(f)$ is a cofibation.

Thus, $F(f)$ has the left lifting property with respect to p, and by the first part of the proof, f has the left lifting property with respect to $U(p)$. Thus, $U(p)$ has the right lifting property with respect to all cofibrations, and by lemma 1.1.18, $U(p)$ is a trivial fibration.

(b) Let F be left Quillen. Then F preserves all trivial cofibraitons. Let p be a cofibration and let f be a trivial cofibration. Since F preserves trivial cofibrations, we have that $F(f)$ is a trivial cofibration. Thus, $F(f)$ has the left lifting property with respect to p, and by the first part of the proof, f has the left lifting property with respect to $U(p)$. Thus, $U(p)$ has the right lifting property with respect to all trivial cofibrations, and by lemma 1.1.18, $U(p)$ is a fibration.

2. ($\Longleftarrow$)

   (a) Let U be right Quillen. Then U preserves all fibrations. Let f be a trivial cofibration and let p be a fibration. Since U preserves fibrations, we have that $U(p)$ is a fibration. Thus, f has the left lifting property with respect to $U(p)$, and by the first part of the proof, $F(f)$ has the left lifting property with respect to p. Thus, $F(f)$ has the left lifting property with respect to all fibrations, and by lemma 1.1.18, $F(f)$ is a trivial cofibration.

   (b) Let U be right Quillen. Then U preserves all trivial fibrations. Let f be a cofibration and let p be a trivial fibration. Since U preserves trivial fibrations, we have that $U(p)$ is a trivial fibration. Thus, f has the left lifting property with respect to $U(p)$, and by the first part of the proof, $F(f)$ has the left lifting property with respect to p. Thus, $F(f)$ has the left lifting property with respect to all trivial fibraitons, and by lemma 1.1.18, $F(f)$ is a cofibration.

■

Remark 3.1.5. This implies that all adjunctions preserve lifts. We spell out the explicit formula incase it is needed in the future. In what follows, let $(F, U, \eta, \varepsilon) : \mathcal{C} \to \mathcal{D}$ be any adjunction (not necessarily between model categories), let $f : c \to c'$ in $\mathcal{C}$, and let $p : d \to d'$ in $\mathcal{D}$.

1. Let $F(f)$ have the left lifting property with respect to p, and let $\ell : F(c') \to d$ be a lift. Then f has the left lifting property with respect to $U(p)$, and the lift ℓ corresponds to a lift $U(\ell) \circ \eta_{c'}$.
2. Let f have the left lifting property with respect to $U(p)$, and let $\ell : c' \to U(d)$ be a lift. Then $F(f)$ has the left lifting property with respect to p, and the lift ℓ corresponds to a lift $\varepsilon_d \circ F(\ell)$.

Remark 3.1.6. Lemma 3.1.4 explains why we only defined left and right variants of Quillen functors, instead of considering a functor that preserves all cofibrations, fibrations, and weak equivalences. Since left and right Quillen functors always come in adjunctions, by lemma 3.1.4, any Quillen functor necessarily has a corresponding adjoint Quillen functor, and thus requiring a functor to be left *and* right Quillen implies that this functor has left and right adjoints, both of which are necessarily Quillen. This is a lot to ask for, and for that reason we rarely see functors that are left and right Quillen. Thus, it is much more natural to define two flavors of Quillen functors.

Remark 3.1.7. It is likely that lemma 3.1.4 can be proven in another way. To simplify notation, note that the arrow category $\operatorname{Map}\mathcal{C}$ is the same as the functor category $\mathcal{C}^2$, where 2 here denotes the obvious poset category. Recall that a lift is really a factorization of a morphism $(h, k) : f \to U(p)$ in $\operatorname{Map}\mathcal{C}$ as

$$p \xrightarrow{(h,1)} \ell \xrightarrow{(1,k)} k,$$

where the intermediate object ℓ is the solution to the lifting problem. If one could promote the adjunction $F \dashv U : \mathcal{C} \to \mathcal{D}$ to an adjunction between the arrow categories $F^2 \dashv U^2 : \mathcal{C}^2 \to \mathcal{D}^2$, then one obtains an isomorphism $\mathcal{D}^2(F^2(f), p) \cong \mathcal{C}^2(f, U^2(p))$ natural in f and p. I am not fully sure where to go from here, although I think one can make use of the factorization by constructing the relevant naturality square where the arrow $\mathcal{C}^2(F^2(f), \ell) \to \mathcal{C}^2(F^2(f), p)$ is given by postcomposing with $(1, k)$, and assuming $(h, 1) \in \mathcal{C}^2(F^2(f), \ell)$ since $F(f)$ has the left lifting property with respect to p. We then follow $(h, 1)$ around the naturality square, and consider whether or not the image is (h, k). I am not sure if this proves the statement, and one would have to do some dirty work with adjunctions, but there should at least be a way to approach the problem from the viewpoint of promoting the adjunctions, even if I am not approaching it properly.

Example 3.1.8. Let $\mathcal{C}$ be a model category, $I \in \mathcal{S}\mathrm{et}$, and let $\mathcal{C}^I$ denote the I-indexed product. The product functor $\times^I : \mathcal{C}^I \to \mathcal{C}$ is of course right-adjoint to the diagonal functor $\Delta : \mathcal{C} \to \mathcal{C}^I$, and by definition of the product model structure as seen in example 1.1.10, clearly Δ preserves cofibrations and trivial cofibrations. Thus, $\Delta \dashv \times^I$ is a Quillen adjunction.

Example 3.1.9. Recall that given a model category $\mathcal{C}$, construction 1.1.13 gives a model structure for the category $\mathcal{C}_* = (* \downarrow \mathcal{C})$. Since we defined arrows as cofibrations, fibrations, or weak equivalences precisely when their underlying arrows are, the forgetful functor $U : \mathcal{C}_* \to \mathcal{C}$ obviously preserves these properties. Since $(-)_+ \dashv U$, we have that U is right Quillen. As noted in remark 1.1.15, the category $(x \downarrow \mathcal{C})$ is a model category for *any* $x \in \mathcal{C}$, and thus the forgetful functor $U : \mathcal{C}_x \to \mathcal{C}$ is always right Quillen, provided a left adjoint exists. The left adjoint should be defined as $a \mapsto a \amalg x$, but I am not fully confident this produces an adjunction in the same way $x \mapsto x \amalg *$ does, and would have to investigate further. If I ever need a slice or coslice category of this form, I will check this.

Construction 3.1.10 (Category of Model Categories)**.** Since Quillen functors come in two flavors, and are always packaged into Quillen adjunctions, it would be most natural to consider the adjunction itself as a morphism between model categories. We review the construction of the 2-category $\mathcal{A}\mathrm{dj}$ whose morphisms are adjunctions, in order to naturally introduce the 2-category $\mathcal{M}\mathrm{od}$ of model categories, Quillen adjunctions, and natural transformations. Let $(F, U, \varphi) : \mathcal{C} \to \mathcal{D}$ and $(F', U', \varphi') : \mathcal{D} \to \mathcal{E}$ be adjunctions. Define the composition adjunction by

$$(F' \circ F, U \circ U' \varphi \circ \varphi') : \mathcal{C} \to \mathcal{E},$$

$$\varphi \circ \varphi' : \mathcal{E}(F'F(a), b)` \xrightarrow{\varphi'} \mathcal{D}(F(a), U'(b)) \xrightarrow{\varphi} \mathcal{E}(a, UU'(b)).$$

Composition of adjunctions is manifestly associative, since functor composition and vertical composition of natural transformations are associative. The identity adjunction $(1, 1, 1_1)$ is clearly an identity for composition. Now given two adjunctions with the same domain and codomain $F \dashv U, F' \dashv U' : \mathcal{C} \to \mathcal{D}$, define the 2-cells as a pair of natural transformations $\alpha : F \Rightarrow F'$ and $\beta : U' \Rightarrow U$, such that α and β are *mates*. The mate condition states that given α, β must be defined as the composite

$$\beta : U' \xrightarrow{\eta U'} UFU' \xrightarrow{U\alpha U'} UF'U' \xrightarrow{U\varepsilon'} U.$$

One may also define α via β by the composite

$$\alpha : F \xrightarrow{F\eta'} FU'F' \xrightarrow{F\beta F'} FUF' \xrightarrow{\varepsilon F'} F'.$$

I am only 70% confident the second formula is correct, so I will attempt to (if it is ever needed) always construct β from α. If a time comes that I cannot do this, I will add a remark detailing how these compositions arise to ensure the bottom composite is correct[1]. Since one mate determines the other, usually only one is specified. Vertical composition is standard vertical composition, and similarly horizontal composites are given by standard horizontal composition.

Now we specialize to model categories and Quillen functors. It is fairly clear that composing left (right) Quillen functors gives a left (right) Quillen functor, so we define the category $\mathcal{M}\mathrm{od}$ of model categories by taking objects to be model categories, morphisms to be Quillen adjunctions, and 2-morphisms to be mate-pairs of natural transformations, as described above.

Remark 3.1.11. Construction 3.1.10 was originally written with the assumption that $\mathcal{M}\mathrm{od}$ would denote the category of *small* model categories. However, after reading the following section in Hovey's text, he notes a problem I did not consider: there are *no nontrivial small model categories*, because *model categories need be complete and cocomplete.* The solution to this foundational problem is apparently constructing $\mathcal{M}\mathrm{od}$ as a 2-category, allowing one to take the collection of objects to form a "superclass", the collection of 1-morphisms to form a class, and the collection of 2-morphisms to also form a class. I have no idea what a superclass is formally, and Hovey only informally states that it is "to a class as a class is to a set". I will thus disregard the foundational problem and assume either someone else has solved it, or ZFC is trash and we should replace it :3

Remark 3.1.12. If $(F, U, \varphi) : \mathcal{C} \to \mathcal{D}$ is a Quillen adjunction, then

$$D(F, U, \varphi) := (U, F, \varphi^{-1}) : D\mathcal{D} \to D\mathcal{C}$$

is also a Quillen adjunction between the dual categories. D is indeed contravariantly functorial, and is a self-inverse.

Proposition 3.1.13. *With notation as in construction 1.1.13, a Quillen adjunction (F, U, φ) : $\mathcal{C} \to \mathcal{D}$ induces a Quillen ajunction $(F_*, U_*, \varphi_*) : \mathcal{C}_* \to \mathcal{D}_*$. Moreover, there are isomorphisms $F_*(x_+) \cong (F(x))_+$ natural in x. The correpsondence $(F, U, \varphi) \mapsto (F_*, U_*, \varphi_*)$ is functorial up to natural isomorphism.*

Proof. Define $U_*(x, u) = (U(x), U(u))$. Since U is right adjoint, it preserves the terminal object, and thus $U(u) : U(x) \to *$; thus U_* is well-defined on objects. The action on morphisms is even more obvious. Let $f : (x, u) \to (y, v)$, meaning in particular that $v = fu$. Define $U_*(f) = U(f)$, and note that $U(f) : U(x) \to U(y)$, and by functoriality, $U(v) = U(fu) = U(f)U(u)$. Thus, $U(f) : (U(x), U(u)) \to (U(y), U(v))$ is well-defined as well. Since $U_* = U$ on morphisms, we immediately have by definition of the model structure on $\mathcal{C}_*$ that U preserves fibrations and trivial fibrations. It thus suffices to construct a left adjoint.

The left adjoint is constructed by pushout diagrams as

$$\begin{tikzcd} F(*) \arrow[r, "F(u)"] \arrow[d] & F(x) \arrow[d] \\ * \arrow[r] & F_*(x, u) \end{tikzcd}$$

Let $f : (x, u) \to (y, v)$ in $\mathcal{C}$. Note that $F(f) \circ F(u) = F(v)$. Factoring the pushout square for

[1]The enterprising reader can look to Hovey 1.4 to see the rule determining the first composite, and can work through the algebra to determine the second composite.

$F_*(y,v)$ along this composite, one obtains by universality of pushout, given the commutative diagram of solid arrows

$$\begin{tikzcd}
F(*) \arrow[r, "F(u)"] \arrow[d] & F(x) \arrow[r, "F(f)"] \arrow[d] & F(y) \arrow[d] \\
* \arrow[r] \arrow[rr, bend right] & F_*(x,u) \arrow[r, dashed, "k"'] & F_*(y,v)
\end{tikzcd}$$

there is a unique dashed arrow rendering the diagram commutative. We define $F_*(f)$ as the dashed arrow k.

Let $f : F_*(x,u) \to (y,v)$ in $\mathcal{D}_*$. By universality of pushout (aka by precomposing along the relevant arrows), the diagram

$$\begin{tikzcd}
F(*) \arrow[r, "F(u)"] \arrow[d] & F(x) \arrow[d] \arrow[ddr, bend left, "k"] & \\
* \arrow[r] \arrow[drr, bend right] & F_*(x,u) \arrow[dr, "f"] & \\
 & & (y,v)
\end{tikzcd}$$

commutes. Universality guarentees that the arrow k is defined uniquely by this choice of f. The composite $k \circ F(u) : F(*) \to (y,v)$ transposes under U to an arrow $\ell : * \to U(y)$. Some work with adjunctions shows that $\ell = v$, and so by commutivity of the diagram k transposes under $F \dashv U$ to an arrow $(x,u) \to U_*(y,v)$. Everything else is obvious from here (aka I didn't feel like working through the rest of this proof since I already see how it goes). ■

3.2 Derived functors and naturality

Definition 3.2.1 (Derived Functors)**.** Let $\mathcal{C}$ and $\mathcal{D}$ be model categories, and let Q and R be the cofibrant and fibrant replacements, respectively, as defined in remark 1.1.16.

1. If $F : \mathcal{C} \to \mathcal{D}$ is left Quillen, define the *total left derived functor* $\mathbf{L}F : \operatorname{Ho}\mathcal{C} \to \operatorname{Ho}\mathcal{D}$ as the composite
$$\operatorname{Ho}\mathcal{C} \xrightarrow{\operatorname{Ho}Q} \operatorname{Ho}\mathcal{C}_c \xrightarrow{\operatorname{Ho}F} \operatorname{Ho}\mathcal{D}.$$
Given a natural transformation $\tau : F \Rightarrow F'$ of left Quillen functors, define the *total derived natural transformation* $\mathbf{L}\tau$ as $\operatorname{Ho}\tau \circ \operatorname{Ho}Q$, which we recall is defined as $(\mathbf{L}\tau)_x = \tau_{Q(x)}$.
2. If $U : \mathcal{D} \to \mathcal{C}$ is right Quillen, define the *total right derived functor* $\mathbf{R}U : \operatorname{Ho}\mathcal{D} \to \operatorname{Ho}\mathcal{C}$ of U to be the composite
$$\operatorname{Ho}\mathcal{D} \xrightarrow{\operatorname{Ho}R} \operatorname{Ho}\mathcal{D}_f \xrightarrow{\operatorname{Ho}U} \operatorname{Ho}\mathcal{C}.$$
Given a natural transformation $\tau : U \Rightarrow U'$ of right Quillen functors, define the *total derived natural transformation* $\mathbf{R}\tau$ as $\operatorname{Ho}\tau \circ \operatorname{Ho}R$, which we recall is defined as $(\mathbf{R}\tau)_x = \tau_{R(x)}$.

♡

Terminology 3.2.2. A functor in practice is almost never left and right Quillen, so we generally leave the left/right implicit and refer only to the functor's total derived functor.

Remark 3.2.3. Note that even if $F : \mathcal{C} \to \mathcal{D}$ is not left Quillen, provided it preserves weak equivalences between cofibrant objects, or by Ken Brown's lemma 1.1.21 provided that it preserves trivial cofibrations betwen cofibrant objects, we still obtain a defintion of $\operatorname{Ho} F$ by universality of $\operatorname{Ho} \mathcal{C}_c$, and thus can still define $\mathbf{L}F$. The dual, of course, applies to total right derived functors.

Remark 3.2.4. Riehl defines total left derived functors between general homotopic categories as the Kan extension $\operatorname{Lan}_\gamma(\delta F)$, where $\delta : \mathcal{D} \to \operatorname{Ho} \mathcal{D}$ and $\gamma : \mathcal{C} \to \operatorname{Ho} \mathcal{C}$ are the localization functors. An immediate thought is then to consider whether or not these definitions are equivalent. I see no obvious way to prove this is true, but I am sure it is. Most likely the proof is fairly complex, so I will return to it later in the text when I have sufficient time.

Remark 3.2.5. Derivation is functorial on functor categories. In other words, given $\tau', \tau : F \Rightarrow G$, we have

$$\mathbf{L}(\tau' \circ \tau)_x = (\tau' \circ \tau)_{Q(x)} = \tau'_{Q(x)} \circ \tau_{Q(x)} = (\mathbf{L}\tau')_x \circ (\mathbf{L}\tau)_x$$

by unraveling definitions. Thus, $\mathbf{L}(\tau' \circ \tau) = \mathbf{L}\tau' \circ \mathbf{L}\tau$, and similarly for $\mathbf{R}$. We also see that $\mathbf{L}1_F = 1_{\mathbf{L}F}$, and thus $\mathbf{L}$ and $\mathbf{R}$ are functorial on functor categories $\mathbf{L}, \mathbf{R} : \mathcal{D}^{\mathcal{C}} \to \operatorname{Ho} \mathcal{D}^{\operatorname{Ho} \mathcal{C}}$.

However, functoriality falls apart when considering functors as arrows, rather than objects. Note that $\mathbf{L}(1_{\mathcal{C}}) = \operatorname{Ho} Q \neq 1$, and there are similar discrepancies that arise. However, passing to a bicategorical structure shows that $\mathbf{L}$ and $\mathbf{R}$ can be viewed as pseudofunctor.

Theorem 3.2.6. *Let $\mathcal{C}$ be a model category. Then there is a natural isomorphism $\alpha : L(1_{\mathcal{C}}) \cong 1_{\operatorname{Ho} \mathcal{C}}$. Additionally, for any pair of left Quillen functors $F : \mathcal{C} \to \mathcal{D}$ and $F' : \mathcal{D} \to \mathcal{E}$, there is a natural isomorphism*

$$m = m_{F',F} : \mathbf{L}F' \circ \mathbf{L}F \cong \mathbf{L}(F' \circ F).$$

These natural isomorphisms satisfy the following coherence properties.

1. *Given $F : \mathcal{C} \to \mathcal{C}'$, $F' : \mathcal{C}' \to \mathcal{C}''$, and $F'' : \mathcal{C}'' \to \mathcal{C}'''$ left Quillen, the following associativity diagram commutes:*

$$\begin{tikzcd}
(\mathbf{L}F'' \circ \mathbf{L}F') \circ \mathbf{L}F \arrow[r, "m_{F'',F'}\mathbf{L}F"] \arrow[d, equal] & \mathbf{L}(F'' \circ F') \circ \mathbf{L}F \arrow[r, "m_{(F''\circ F'),F}"] & \mathbf{L}((F'' \circ F') \circ F) \arrow[d, equal] \\
\mathbf{L}F'' \circ (\mathbf{L}F' \circ \mathbf{L}F) \arrow[r, "\mathbf{L}F'' m_{F',F}"'] & \mathbf{L}F'' \circ \mathbf{L}(F' \circ F) \arrow[r, "m_{F'',(F'\circ F)}"'] & \mathbf{L}(F'' \circ (F' \circ F))
\end{tikzcd}$$

2. *If $F : \mathcal{C} \to \mathcal{D}$ is left Quillen, the following left unit coherence diagram commutes.*

$$\begin{tikzcd}
\mathbf{L}1_{\mathcal{D}} \circ \mathbf{L}F \arrow[r, "m"] \arrow[d, "\alpha\mathbf{L}F"'] & \mathbf{L}(1_{\mathcal{D}} \circ F) \arrow[d, equal] \\
1_{\operatorname{Ho} \mathcal{D}} \circ \mathbf{L}F \arrow[r, equal] & \mathbf{L}F
\end{tikzcd}$$

3. *If $F : \mathcal{C} \to \mathcal{D}$ is left Quillen, the following right unit coherence diagram commutes.*

$$\begin{tikzcd}
\mathbf{L}F \circ \mathbf{L}1_{\mathcal{C}} \arrow[r, "m"] \arrow[d, "\mathbf{L}F\alpha"'] & \mathbf{L}(F \circ 1_{\mathcal{C}}) \arrow[d, equal] \\
\mathbf{L}F \circ 1_{\operatorname{Ho}\mathcal{C}} \arrow[r, equal] & \mathbf{L}F
\end{tikzcd}$$

Proof. I hate coherence checks. Define $\alpha : \mathbf{L}(1_{\mathcal{C}}) \Rightarrow 1_{\operatorname{Ho}\mathcal{C}}$ as $\operatorname{Ho} q$, where we recall from 1.1.16 that $q : Q \Rightarrow 1_{\mathcal{C}}$ is a natural trivial fibration, and the homotopy category inverts components of q, making $\operatorname{Ho} q$ is a natural isomorphism. Define m as

$$m_{F',F} : (\mathbf{L}F' \circ \mathbf{L}F)(x) = F'QFQ(x) \xrightarrow{F'qFQ(x)} F'FQ(x) = \mathbf{L}(F'F)(x).$$

Thus $m_{F',F}$ is natural in x by definition. Note that this definition involves a serious abuse of notation, since the total derived functor is not FQ, but $\operatorname{Ho} F \operatorname{Ho} Q$. They are, however, equivalent on objects, so we can simply lower m to the homotopy categories. Since $Q(x)$ is cofibrant, and since F, F', Q preserve cofibrant objects, we have that $F'qFQ(x)$ is an arrow between cofibrant objects. Note by remark 3.2.7 that Q preserves weak equivalences, and thus both the source and target of $m_{F',F}$ preserve weak equivalences. Thus, $m_{F',F}$ lowers to a natural transformation in $\operatorname{Ho}\mathcal{C}$ by universality. Since q is a trivial fibration, and in particular a weak equivalence, and since F preserves cofibrant objects, we have that $q_{FQ(x)} : QFQ(x) \to FQ(x)$ is a weak equivalence between cofibrant objects. By Ken Brown's lemma 1.1.21, since F' preserves trivial cofibrations between cofibrant objects, it necessarily preserves weak equivalences between them, and thus $(m_{F',F})_x = F'(q_{FQ(x)})$ is a weak equivalence. Thus, this lowers to an isomorphism in $\operatorname{Ho}\mathcal{C}$.

Now we show the associativity coherence diagram commutes. It suffices to show

$$(F''F'q_{FQ(x)})c \circ (F''q_{F'QFQ(x)}) = (F''q_{F'FQ(x)}) \circ (F''QF'q_{FQ(x)})$$

as maps $F''QF'QFQ(x) \Rightarrow F''F'FQ(x)$[a]. "This follows by naturality of q." I don't like coherence checks so I agree! Note that the left unit coherence diagram expresses the equality

$$1_{\mathcal{D}}qFQ = qFQ,$$

where we once again drop the Ho and write $\mathbf{L}F$ as FQ, with the understanding that mapping under Ho gives us the proper definition between the homotopy categories.

The right coherence diagram depicts the equality

$$FqQ = FQq : FQQ \Rightarrow FQ,$$

which does not hold in $\mathcal{C}$ in general. We must thus transfer to $\operatorname{Ho}\mathcal{C}$ to show this statement. To do this, we show the components are equal at components x for x cofibrant, since all objects in $\operatorname{Ho}\mathcal{C}$ are isomorphic to a cofibrant object (theorem 1.2.18). Naturality implies $q_x \circ q_{Q(x)} = q_x \circ Q(q_x)$. Since q_x is a weak equivalence between cofibrant objects, since x is cofibrant, $F(q_x)$ is also a weak equivalence. Thus, $F(q_{Q(x)}) == FQ(q_x)$ in $\operatorname{Ho}\mathcal{C}$, since one can take F on both sides of the above equality then cancel $F(q_x)$, since it is inverted in $\operatorname{Ho}\mathcal{C}$. ■

[a] $FQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQFQ$

Remark 3.2.7. We still have to show Q preserves weak equivalences, and dually so does R. I'm suprised I haven't shown this earlier in my notes. Recall $Q(x)$ is defined via the functorial factorization

$$0 \xrightarrow{\alpha(0\to x)} Q(x) \xrightarrow{\beta(0\to x)} x,$$

where as usual (α, β) denotes our cofibration/trivial fibration factorization. Given a weak equivalence $f : x \to y$, remark 1.1.2 factors the diagram as

$$\begin{tikzcd} 0 \arrow[r, "\alpha(0\to x)"] \arrow[d, equal] & Q(x) \arrow[r, "\beta(0\to x)"] \arrow[d, "Q(f)"'] & x \arrow[d, "f"] \\ 0 \arrow[r, "\alpha(0\to y)"'] & Q(y) \arrow[r, "\beta(0\to y)"'] & y \end{tikzcd}$$

Note that since $\beta(0 \to x)$ is a trivial fibration, and since f is a weak equivalence, their composite $f \circ \beta(0 \to x)$ is a weak equivalence. By commutivity, the composite $\beta(0 \to y) \circ Q(f)$ is a weak equivalence, and since $\beta(0 \to y)$ is a trivial fibration, by 2-out-of-3 we have that $Q(f)$ is a weak equivalence.

Lemma 3.2.8. *Let $\sigma : F \Rightarrow G$ be a natural transformation of weak left Quillen functors $\mathcal{C} \to \mathcal{D}$, and let $\tau : F' \Rightarrow G'$ be a natural transformation of weak left Quillen functors $\mathcal{D} \to \mathcal{E}$. Let m be as defined in theorem 3.2.6. Then the diagram*

$$\begin{tikzcd} \mathbf{L}F' \circ \mathbf{L}F \arrow[r, "m"] \arrow[d, "\mathbf{L}\tau\cdot\mathbf{L}\sigma"'] & \mathbf{L}(F' \circ F) \arrow[d, "\mathbf{L}(\tau\cdot\sigma)"] \\ \mathbf{L}G' \circ \mathbf{L}G \arrow[r, "m"'] & \mathbf{L}(G' \circ G) \end{tikzcd}$$

where we denote horizontal composition by $\cdot$ for emphasis. In particular, $\mathbf{L}$ preserves horizontal composites up to isomorphism.

Proof. Recall the horizontal composite is defined via

$$(\tau \cdot \sigma)_x = \tau_{G(x)} \circ F'(\sigma_x) = G'(\sigma_x) \circ \tau_{F(x)}.$$

Unraveling definitions,

$$\mathbf{L}(\tau \cdot \sigma)_x \circ m = \tau_{GQ(x)} \circ F'(\sigma_{Q(x)},$$

where we continue to not specify the Ho for simplicity, since the definition of Ho on natural transformations is so similar. Naturality of q rewrites this composite as $\tau_{GQ(x)} \circ F'(q_{GQ(x)}) \circ F'Q(\sigma_{Q(x)})$. Naturality of τ then rewrites this composite as $G'(q_{GQ(x)}) \circ \tau_{QGQ(x)} \circ F'Q(\sigma_{Q(x)})$, which is the definition of $m \circ (\mathbf{L}\tau \cdot \mathbf{L}\sigma)$. ■

Remark 3.2.9. The theory here is fairly dry, we are just kind of proving some basic things work and are functors, so I hope any reader will forgive me for skipping over some steps and at times basically just paraphrasing Hovey.

Remark 3.2.10. The statements of lemma 3.2.8 and theorem 3.2.6 can be summarized as stating that $\mathbf{L}$ is a psuedofunctor from the 2-category of model categories with morphisms given by *left Quillen functors* to the 2-category $\mathcal{C}\mathrm{at}$. Of course the dual of this is that $\mathbf{R}$ is a pseudofunctor from the 2-category of model categories with morphisms as *right* Quillen functors to $\mathcal{C}\mathrm{at}$. These definitions differ from our definition of $\mathcal{M}\mathrm{od}$ in construction 3.1.10, so we would like a way to make the same claim for adjuntions. To do this, we introduce the notion of a *derived adjunction*.

Lemma 3.2.11. *Let $(F, U, \varphi) : \mathcal{C} \to \mathcal{D}$ be a Quillen adjunction. Then $\mathbf{L}F$ and $\mathbf{R}U$ assemble into an adjunction $\mathbf{L}(F, U, \varphi) \coloneqq (\mathbf{L}F, \mathbf{R}U, \mathbf{R}\varphi)$, called the* derived adjunction.

Proof. It suffices to construct $\mathbf{R}\varphi$ as an isomorphism natural in x and y. Recall by theorem 1.2.18 that there is a natural isomorphism $\operatorname{Ho}\mathcal{D}(FQ(x), y) \cong \mathcal{D}(FQ(x), R(y))/{\sim}$, since $Q(x)$ is cofibrant. Although this is not explicitly stated, applying only the natural transformation $1 \Rightarrow R$ in the proof easily shows this to be the case. Similarly, we have a natural isomorphism $\operatorname{Ho}\mathcal{C}(x, UR(y)) \cong \mathcal{C}(Q(x), UR(y))/{\sim}$. Since the object function of $\mathbf{L}F$ and $\mathbf{R}U$ are the same as FQ and UR, it suffices to show the middle isomorphism (marked with a ?) in the chain

$$\operatorname{Ho}\mathcal{C}(x, UR(y)) \cong \mathcal{C}(Q(x), UR(y))/{\sim} \overset{?}{\cong} \mathcal{D}(FQ(x), R(y))/{\sim} \cong \operatorname{Ho}\mathcal{D}(FQ(x), y)$$

indeed exists, and is natural. Without the quotients, this isomorphism is just a special case of φ, so we need only show φ respects the isomorphism relation, and then to define $\mathbf{R}\varphi$ by composing along the given isomorphisms.

Since $Q(x)$ and $R(y)$ are in particular cofibrant and fibrant, respectively, we show that φ respects homotopy between cofibrant and fibrant objects. Let $f, g : F(a) \to b$ be homotopic maps with a cofibrant and b fibrant. Let b' be a path object for b, and $H : F(a) \to b'$ a right homotopy from f to g. Since U is right Quillen, it preserves products, fibrations, and weak equivalences between fibrant objects. In particular, $U(b')$ is a path object for $U(b)$, since the product, weak equivalence, and fibration are preserved. To illustrate this in more precision, U applied to the path object gives

$$U(b) \longrightarrow U(b') \longrightarrow U(b \times b).$$

The left arrow is a weak equivalence because b is fibrant, and we can choose b' to be fibrant by proposition 1.2.12 allowing arbitrary choices of path objects, and remark 1.2.11 giving the relevant path object; U then preserves weak equivaleces between fibrant objects. The right arrow is a fibration since U preserves them, and since products are preserved, $U(b \times b) \cong U(b) \times U(b)$; isomorphisms are trivial fibrations so we thus keep our path object.

The homotopy diagram is given as

$$\begin{tikzcd}
& & b \\
F(a) \arrow[r, "H"] \arrow[urr, bend left, "f"] \arrow[drr, bend right, "g"'] & b' \arrow[r, "{(p_0,p_1)}"] \arrow[ur, "p_0"] \arrow[dr, "p_1"'] & b \times b \arrow[u] \arrow[d] \\
& & b
\end{tikzcd}$$

Note here that postcomposing H by p_0 can be viewed as $\mathcal{D}(F(1_a), p_0)(H)$. Naturality of φ gives the outer square

$$\begin{tikzcd}
\mathcal{D}(F(a), b') \arrow[rr, "\varphi_{a,b'}"] \arrow[dd, "\mathcal{D}(F(1_a),p_0)"', "(p_0)_*"] & & \mathcal{C}(a, U(b')) \arrow[dd, "U(p_0)_*"', "\mathcal{C}(1_a,U(p_0))"] \\
& \begin{matrix} H & \longmapsto & \varphi(H) \\ \downarrow & & \downarrow \\ p_0H = f & \longmapsto & \varphi(f) = U(p_0)\varphi(H) \end{matrix} & \\
\mathcal{D}(F(a), b) \arrow[rr, "\varphi_{a,b}"'] & & \mathcal{C}(a, U(b))
\end{tikzcd}$$

Following H around the outer square, as indicated by the inner square, expresses the equality $\varphi(f) = U(p_0)\varphi(H)$. Replacing p_0 with p_1 gives the equality $\varphi(g) = U(p_1)\varphi(H)$. Thus we obtain that the diagram

$$\begin{tikzcd}
& & & U(b) \\
a \arrow[r, "\varphi(H)"] \arrow[urrr, bend left, "\varphi(f)"] \arrow[drrr, bend right, "\varphi(g)"'] & U(b') \arrow[urr, "U(p_0)"] \arrow[drr, "U(p_1)"'] \arrow[r, "U(p_0{,}p_1)", "(U(p_0){,}U(p_1))"'] & U(b \times b) \cong U(b) \times U(b) \arrow[ur] \arrow[dr] & \\
& & & U(b)
\end{tikzcd}$$

commutes. Since $U(b')$ is a path object for $U(b)$, we have that $\varphi(H)$ is a (right) homotopy from $\varphi(f)$ to $\varphi(g)$. Since b is fibrant and U is right Quillen, so is $U(b)$. Since a is cofibrant, by proposition 1.2.12 gives us $\varphi(f) \sim \varphi(g)$.

We must also show the converse; that is, we must also show φ^{-1} respects homotopy. Let $f, g : a \to U(b)$ be morphisms with a cofibrant and b fibrant; let

$$a \amalg a \xrightarrow{i_0+i_1} a' \longrightarrow a$$

be a cylinder object for a, where we once again choose a' cofibrant by remark 1.2.11 and proposition 1.2.12; and let

$$\begin{tikzcd}
a \arrow[d] \arrow[dr, "i_0"] \arrow[drr, bend left, "f"] & & \\
a \amalg a \arrow[r, "i_0+i_1"] & a' \arrow[r, "H"] & U(b) \\
a \arrow[u] \arrow[ur, "i_1"'] \arrow[urr, bend right, "g"'] & &
\end{tikzcd}$$

be a left homotopy $f \sim g$. Naturality of φ^{-1} gives the outer square

$$\begin{tikzcd}
\mathcal{C}(a',U(b)) \arrow[rrr, "\varphi^{-1}_{a',b}"] \arrow[ddd, "\mathcal{C}(i_0,U(1_b))"', "(i_0)^*"] & & & \mathcal{D}(F(a'),b) \arrow[ddd, "F(i_0)^*"', "\mathcal{D}(F(i_0),1_b)"] \\
& H \arrow[r, mapsto] \arrow[d, mapsto] & \varphi^{-1}(H) \arrow[d, mapsto] & \\
& Hi_0 = f \arrow[r, mapsto] & \varphi^{-1}(f) = \varphi^{-1}(H)F(i_0) & \\
\mathcal{C}(a,U(b)) \arrow[rrr, "\varphi^{-1}_{a,b}"'] & & & \mathcal{D}(F(a),b)
\end{tikzcd}$$

Once again, tracing H along the outer square, as indicated by the inner square, depicts the equality $\varphi^{-1}(f) = \varphi^{-1}(H)F(i_0)$, and constructing the same square for g gives the equality $\varphi^{-1}(g) = \varphi^{-1}(H)F(i_1)$. We thus obtain that the diagram

$$\begin{tikzcd}
F(a) \arrow[d] \arrow[dr, "F(i_0)"] \arrow[drr, bend left, "\varphi^{-1}(f)"] & & \\
F(a) \amalg F(a) \arrow[r, "F(i_0)+F(i_1)"] & F(a') \arrow[r, "\varphi^{-1}(H)"] & b \\
F(a) \arrow[u] \arrow[ur, "F(i_1)"'] \arrow[urr, bend right, "\varphi^{-1}(g)"'] & &
\end{tikzcd}$$

commutes by definition of coproduct. It thus suffices to show

$$F(a) \amalg F(a) \cong F(a \amalg a) \longrightarrow F(a') \longrightarrow F(a)$$

is a cylinder object for $F(a)$, and indeed since we chose a' cofibrant, the weak equivalence $F(a') \to F(a)$ is preserved by Ken Brown's lemma 1.1.21, the isomorphism indicated on the left exists, and $F(i_0 + i_1) \cong F(i_0) + F(i_1)$ is a cofibration since F is left Quillen. ■

Example 3.2.12. Let I be a set, and $\mathcal{C}$ a model category. We saw in example 3.1.8 that there is a Quillen adjunction $\Delta \dashv \times^I : \mathcal{C} \to \mathcal{C}^I$, where $\times^I$ is the I-ary product multifunctor, for lack of a better notation[2]. Since $\mathbf{L}\Delta(x) = \Delta Q(x)$, under the isomorphism $\operatorname{Ho}\mathcal{C}^I \cong (\operatorname{Ho}\mathcal{C})^I$ we obtain that $\mathbf{L}\Delta$ is naturally isomorphic to the diagonal functor $\Delta : \operatorname{Ho}\mathcal{C} \to (\operatorname{Ho}\mathcal{C})^I$. Thus, the total right derived functor of a product functor on $\mathcal{C}$ is a product functor on $\operatorname{Ho}\mathcal{C}$. Similarly, we obtain the total left derived functor of a coproduct functor is a coproduct functor. One obtains from this that $\operatorname{Ho}\mathcal{C}$ has small products and coproducts, but we will show even more throughout the text. This is yet another sign that the homotopy category of a model category is fantastically well-behaved.

This was the dullest section of the book I've read, I can't want for Quillen equivalences they look so much cooler. Also why do I keep thinking the plural of equivalences is equivali?

[2] I thought for a short time that this was the *iterated* product, which could be denoted by exponentiation, and I don't know why I thought this since there are about 30 reasons it isn't that. For example, exponentiation is a functor $\mathcal{C} \to \mathcal{C}$, so domains don't match.

3.3 Quillen equivalences

The goal of Quillen equivalences is to establish a condition weaker than that of isomorphism, wherein one may consider two different categories to have the same homotopy theory. Since the homotopy category $\operatorname{Ho}\mathcal{C}$ encodes a lot of the homotopic data of a model category, one is naturally lead to consider Quillen adjunctions $F \dashv U$ where the derived adjunction is an equivalence of categories.

Definition 3.3.1 (Quillen Equivalence)**.** A Quillen adjunction $(F, U, \varphi) : \mathcal{C} \to \mathcal{D}$ is a *Quillen equivalence* if and only if for all cofibrant $x \in \mathcal{C}$ and all fibrant $y \in \mathcal{D}$, a morphism $f : F(x) \to y$ is a weak equivalence in $\mathcal{D}$ if and only if $\varphi(f) : x \to U(y)$ is a weak equivalence in $\mathcal{C}$. More concisely, the adjunction φ preserves weak equivalences, and so does its inverse. ♡

Proposition 3.3.2. *Let $(F, U, \varphi) : \mathcal{C} \to \mathcal{D}$ be a Quillen adjunction. Then the following conditions are equivalence.*

1. *(F, U, φ) is a Quillen equivalence as defined in definition 3.3.1;*
2. *The composition $x \xrightarrow{\eta} UF(x) \xrightarrow{U(r_{F(x)})} URF(x)$ is a weak equivalence for all cofibrant x, and the composition $FQU(x) \xrightarrow{F(q_{U(x)})} FQ(x) \xrightarrow{\varepsilon} x$ is a weak equivalence for all fibrant x;*
3. *$\mathbf{L}(F, U, \varphi)$ as defined in lemma 3.2.11 is an adjoint equivalence of categories.*

Proof. I'm going to start indicating when I've done a proof largely by myself, or when it follows the proof of some other source. This proof is longer, so we follow Hovey's Proposition 1.3.13 for the proof, but of course adding in extra details where needed.

(1 $\implies$ 2) Let (F, U, φ) be a Quillen equivalence, and let x be cofibrant. Note that U preserves fibrant objects and $RF(x)$ is fibrant by remark 1.1.16, and thus $RF(x)$ is fibrant. Note that $r_{F(x)} : F(x) \to RF(x)$ is a trivial fibration by remark 1.1.16; in particular it is a weak equivalence. Applying definition 3.3.1 shows that φ preserves weak equivalences of this form, giving us that $\varphi(r_{F(x)}) : x \to URF(x)$ is a weak equivalence. It suffices to show $\varphi(r_{F(x)}) = U(r_{F(x)}) \circ \eta_x$.

Note that $\mathcal{D}(F(1_x), r_{F(x)}) = (r_{F(x)})_*$ is just postcomposition, and similarly $\mathcal{C}(1_x, U(r_{F(x)})) = U(r_{F(x)})_*$ is also just postcomposition. Thus, naturality of φ gives that the left diagram

$$\begin{tikzcd}
\mathcal{D}(F(x), F(x)) \arrow[r, "\varphi"] \arrow[d, "(r_{F(x)})_*"'] & \mathcal{C}(x, UF(x)) \arrow[d, "U(r_{F(x)})_*"] \\
\mathcal{D}(F(x), RF(x)) \arrow[r, "\varphi"'] & \mathcal{C}(x, URF(x))
\end{tikzcd}
\qquad
\begin{tikzcd}
1_{F(x)} \arrow[r, mapsto] \arrow[d, mapsto] & \eta_x \arrow[d, mapsto] \\
r_{F(x)} \arrow[r, mapsto] & \varphi(r_{F(x)}) = U(r_{F(x)}) \circ \eta_x
\end{tikzcd}$$

commutes. Recall that η_x is defined as $\varphi(1_{F(x)})$. Tracing $1_{F(x)}$ around the left square, as indicated by the right square, yields the equality $U(r_{F(x)}) \circ \eta_x = \varphi(r_{F(x)})$. We showed above that $\varphi(r_{F(x)})$ is a weak equivalence, and thus $U(r_{F(x)}) \circ \eta_x$ is a weak equivalence. The statement for F, fibrant objects, and ε is proven analogously.

(2 $\implies$ 1) For the converse, let $f : F(x) \to y$ be a weak equivalence with x cofibrant and y fibrant. We see by examining the above proof, as indicated more clearly in remark 3.3.3, that $\varphi(f) = U(f) \circ \eta_x$. Naturality of $r : 1 \Rightarrow R$ and functoriality of U give the commutative square on

the left of the diagram

$$\begin{tikzcd}
x \arrow[r, "\eta_x"] \arrow[d, equal] & UF(x) \arrow[r, "U(f)"] \arrow[d, "U(r_{F(x)})"'] & U(y) \arrow[d, "U(r_y)"] \\
x \arrow[r, "U(r_{F(x)})\eta_x"'] & URF(x) \arrow[r, "UR(f)"'] & UR(y)
\end{tikzcd}$$

commutes, and the square on the right commutes manifestly. By remark 3.2.7, R preserves weak equivalences, and thus $R(f)$ is a weak equivalence. Since U is right Quillen, by Ken Brown's lemma 1.1.21 it preserves weak equivalences between fibrant objects, and since $RF(x), R(y)$ are fibrant, $UR(f)$ is a weak equivalence. By hypothesis, $U(r_{F(x)}) \circ \eta_x$ is a weak equivalence, and thus the entire bottom composite is a weak equivalence. Since $r_y : y \to R(y)$ is a trivial cofibration between fibrant objects, $U(r_y)$ is a weak equivalence. Thus, $U(r_y) \circ U(f) \circ \eta$ is a weak equivalence by commutivity of the diagram, and by the 2-out-of-3 property, $U(f) \circ \eta_x = \varphi(f)$ is a weak equivalence. The converse direction is once again the same proof, but using the decomposition $\varphi^{-1}(f) = \varepsilon_x \circ F(f)$, and substituting $\varphi(f)$ for f.

($2 \iff 3$) Recall that an adjoint equivalence of categories is an adjunction where the unit and counit are isomorphisms. Thus, it suffices to show each $\mathbf{R}\eta_x$ and $\mathbf{R}\varepsilon_x$ has an inverse if and only if (2) holds. Investigation of theorem 1.2.18 gives that the unit of the derived adjunction is given by the composite

$$x \xrightarrow{q_x^{-1}} Q(x) \xrightarrow{\eta_x} UFQ(x) \xrightarrow{U(r_{FQ(x)})} URFQ(x).$$

Since q_x^{-1} is an isomorphism, we have that the unit is an isomorphism if and only if $U(r_{FQ(x)}) \circ \eta_x$ is an isomorphism. It suffices to show this is true if and only if $U(r_{F(x)}) \circ \eta_x$ is a weak equivalence for all cofibrant x. In the below diagram, the left square commutes by naturality of η. The right square commutes by functoriality of U and naturality of r.

$$\begin{tikzcd}
Q(x) \arrow[r, "\eta_{Q(x)}"] \arrow[d, "q_x"'] & UFQ(x) \arrow[r, "U(r_{FQ(x)})"] \arrow[d, "UF(q_x)"'] & URFQ(x) \arrow[d, "URF(q_x)"] \\
x \arrow[r, "\eta_x"'] & UF(x) \arrow[r, "U(r_x)"'] & URF(x)
\end{tikzcd}$$

Hence we are showing the top composite is an isomorphism if and only if the bottom is a weak equivalence. Since it should be an isomorphism in the homtopy category, that would mean showing it is a weak equivalence. Indeed, q_x is a weak equivalence, F preserves weak equivalences between the cofibrant objects $Q(x)$ and x, R preserves weak equivalences, and then U preserves weak equivalences between fibrant objects. Thus $URF(q_x)$ is a weak equivalence, and the 2-out-of-3 axiom gives the required iff condition. The converse direction is just the opposite proof. ■

Remark 3.3.3. We see from the first part of the proof of proposition 3.3.2 that for any $f : F(x) \to y$, we have $\varphi(f) = U(f) \circ \eta_x$. Indeed, consider the naturality diagram for φ given by

$$\begin{tikzcd}
\mathcal{D}(F(x), F(x)) \arrow[r, "\varphi"] \arrow[d, "\mathcal{D}(F(1_x), f)"', "f_*"] & \mathcal{C}(x, UF(x)) \arrow[d, "U(f)_*"', "\mathcal{C}(1_x, U(f))"] & 1_{F(x)} \arrow[r, mapsto] \arrow[d, mapsto] & \eta_x \arrow[d, mapsto] \\
\mathcal{D}(F(x), y) \arrow[r, "\varphi"'] & \mathcal{C}(x, U(y)) & f \arrow[r, mapsto] & \varphi(f) = U(f) \circ \eta_x
\end{tikzcd}$$

where we have simply replaced $r_{F(x)}$ by f in the relevant diagram in the above proof. The right square indicates the equality given by following $1_{F(x)}$ around the diagram. We see also by duality that any $f : x \to U(y)$ has $\varphi^{-1}(f) = \varepsilon_x \circ F(f)$.

Corollary 3.3.4. *If (F, U, φ) and (F, U', φ') are Quillen adjunctions $\mathcal{C} \to \mathcal{D}$, then (F, U, φ) is a Quillen equivalence if and only if (F, U', φ') is a Quillen equivalence. Dually, if (F', U, φ'') is another Quillen adjunction, then (F, U, φ) is a Quillen equivalence if and only if (F', U, φ'') is.*

Proof. By proposition 3.3.2, (F, U, φ) is a Quillen equivalence if and only if the derived adjunction $\mathbf{L}(F, U, \varphi)$ is an equivalence of categories, which holds if and only if $\mathbf{L}F$ is an equivalence of categories $\operatorname{Ho}\mathcal{D} \simeq \operatorname{Ho}\mathcal{C}$. This solves the proof, since the implication is bidirectional: adjoint equivalence implies $\mathbf{L}F$ equivalence of categories implies any adjunction containing $\mathbf{L}F$ as the left adjoint is an adjoint equivalence. The other direction is the same. ■

Terminology 3.3.5. By corollary 3.3.4, we often refer to a Quillen equivalence $F \dashv U$ by simply naming one functor in the adjunction.

Corollary 3.3.6. *Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{E}$ be either both left, or both right Quillen functors. Then if two out of the three F, G, GF are Quillen equivalences, so is th third. In particular, left and right Quillen functors satisfy the 2-out-of-3 rule.*

Proof. $\mathbf{L}(GF)$ is naturally isomorphic to $\mathbf{L}G \circ \mathbf{L}F$ by the annoying coherence check proof theorem 3.2.6. It is at this point clear how the theroem concludes: since adjunctions can be composed in the style of construction 3.1.10, since an adjunction is an adjoint equivalence if and only if an adjoint functor is an equivalence of categories, it suffices to show equivalences of categories satisfy the 2-out-of-3 rule. This is a simple check; it's extemely similar to the familiar proof that isomorphisms have a unique inverse, but with some extra horizontal compositions thrown in to keep track of (because the inverse of an equivalence of categories is only unique up to isomorphism). ■

Remark 3.3.7. Corollary 3.3.6 clearly implies the 2-of-3 property holds for $\mathcal{M}\mathrm{od}$ as constructed in construction 3.1.10 with adjunctions as morphisms. The obvious next idea is whether we can construct a model structure on $\mathcal{M}\mathrm{od}$. By the time of Hovey's book, this had not yet been accomplished. I have been unable to find such a model structure in the modern literature, but I did find an expository paper on homotopy limits over diagrams of model categories (arXiv:2411.18546, by Julia E Bergner). The paper states "there is no known 'model category of model categories"'. Although the paper doesn't have citations and I'm not familiar with the researcher, it's published so I don't really doubt it, and I also doubt that if a suitable model structure existed, a reseacher who's been working in the field for two decades would write an expository paper on homotopy limits of model categories *without* figuring out the model structure exists.

Much of this is irrelevant, however. The modern understanding of model categories is as a presentation for $(\infty, 1)$-categories (usually just called ∞-categories), a much more suitable environment for homotopy theory. An ∞-category of ∞-categories $\mathcal{C}\mathrm{at}_\infty$ does indeed exist, and the construction can be readily found in the modern literature. For example, see HTT Defintion 3.0.0.1 (by Jacob Lurie, of course). I'm sure *some* nontrivial model structure exists, in fact I think I could probably brute force one given a few days. However, the question is not whether a model structure exists, but whether it is the *proper* model structure, in that the homotopy theory it generates behaves as expected. In this regard, the ∞-categorical approach is better behaved and better suited for the job.

Remark 3.3.8. A functor $F : \mathcal{C} \to \mathcal{D}$ *reflects* a property P of morphisms if for all $f : x \to y$ in $\mathcal{C}$, $P(F(f)) \implies P(f)$. In particular, F reflects weak equivalences between cofibrant objects if for all $f : x \to y$ between cofibrant objects, whenever $F(f)$ is a weak equivalence, so is f.

Corollary 3.3.9. *Let $(F, U, \varphi) : \mathcal{C} \to \mathcal{D}$ be a Quillen adjunction. The following are equivalence.*

1. *(F, U, φ) is a Quillen equivalence.*
2. *F reflects weak equivalences betwen cofibrant objects and, for every fibrant y, the map $FQU(y) \to y$ is a weak equivalence.*
3. *U reflects weak equivalences between fibrant objects and, for every cofibrant x, the map $x \to URF(x)$ is a weak equivalence.*

The arrows of (2,3) are defined as the composites in proposition 3.3.2 (2).

Proof. We follow the proof of Hovey's corollary 1.3.16. Note errata 6 on the nLab.

$(1 \implies 2, 3)$ Let F be a Quillen equivalence. By proposition 3.3.2 the map $x \to URF(x)$ is a weak equivalence for all cofibrant x and the map $FQU(y) \to y$ is a weak equivalence for all fibrant y. Suppose $f : x \to y$ is a map between cofibrant objects and $F(f)$ is a weak equivalence. Naturality of $q : Q \Rightarrow 1$ gives the commutative diagram

$$\begin{array}{ccc} Q(x) & \xrightarrow{q_x} & x \\ {\scriptstyle Q(f)}\downarrow & & \downarrow{\scriptstyle f} \\ Q(y) & \xrightarrow[q_y]{} & y \end{array}$$

Note here that all objects in the diagram are cofibrant, and since components of q are weak equivalences, by Ken Brown's lemma 1.1.21 F preserves the weak equivalences q_x, q_y. In particular, in the commutative diagram

$$\begin{array}{ccc} FQ(x) & \xrightarrow{F(q_x)} & F(x) \\ {\scriptstyle FQ(f)}\downarrow & & \downarrow{\scriptstyle F(f)} \\ FQ(y) & \xrightarrow[F(q_y)]{} & F(y) \end{array}$$

the rows are weak equivalences, and by assumption $F(f)$ is a weak equivalence. By the 2-out-of-3 rule, $FQ(f)$ is a weak equivalence. Lowering to the homotopy category, $\mathbf{L}F(f)$ is thus an isomorphism. Since $\mathbf{L}F$ is an equivalence of categories by proposition 3.3.2, it is easily shown that f is an isomorphism in the homotopy category, and by theorem 1.2.18 f is a weak equivalence. The dual argument proves U reflects weak equivalences between fibrant objects, so we have $(1 \implies 2, 3)$.

$(2 \implies 1)$ By proposition 3.3.2, it suffices to show $\mathbf{L}(F, U, \varphi)$ is an equivalence of categories. I didn't feel like typing this part up lmfao ■

Proposition 3.3.10. *Let $F : \mathcal{C} \to \mathcal{D}$ be a Quillen equivalence, and suppose the terminal object $* \in \mathcal{C}$ is cofibrant and that F preserves the terminal object. Then $F_* : \mathcal{C}_* \to \mathcal{D}_*$ as constructed in proposition 3.1.13 is a Quillen equivalence.*

Proof. Let $F \dashv U$, and recall $U(x,v) = (U(x), U(v))$ and $U_*(f) = U(f)$. By corollary 3.3.9, since U reflects weak equivalences between fibrant objects, very clearly so does U_*. By corollary 3.3.9, it suffices to show for any cofibrant $(x,v) \in \mathcal{C}_*$, the map $U(r_{F(x)}) \circ \eta_{(x,v)} : (x,v) \to U_* R F_*(x,v)$ is a weak equivalence. Note that $(*,1)$ is initial in $\mathcal{C}_*$, since $u : (*,1) \to (x,v)$ must preserve basepoints, and thus $u = v$. Thus by construction 1.1.13, (x,v) is cofibrant if and only if $v : * \to x$ is a cofibration in $\mathcal{C}$.

Let (x,v) be cofibrant. Then since $0 \to *$ is a cofibration and $* \to x$ is a cofibration, the composite

$$\begin{tikzcd} 0 \arrow[r] \arrow[rr, bend left] & * \arrow[r] & x \end{tikzcd}$$

is a cofibration. Thus, x is cofibrant as an object of $\mathcal{C}$. Let $V : \mathcal{C}_* \to \mathcal{C}$ denote the forgetful functor. Since $VU_* = UV$, and since V preserves weak equivalences, it suffices to show the map $x \to VU_*RF_*(x) = UVRF_*(x,v)$ is a weak equivalence, since U reflects weak equivalences. The fibrant replacement R on $\mathcal{D}_*$ is the fibrant replacement of $\mathcal{D}$ together with the included basepoint, so $UVRF_*(x,v) = URVF_*(x)$. Since $F(*) \cong *$, investigation of the relevant pushout diagram gives $F_*(x,v) \cong (F(x), F(v))$. A diagram chase I am not interested in doing gives the result. ■

Remark 3.3.11. I have struggled to get through this section in a timely manner due to other responsibilities, which is burning me out slightly and I am not interested in completing the proofs with full rigor. Luckily we are at the end of the section, so I hope the reader will forgive me. Tomorrow we move on to a new topic, which may be more interesting.

Lemma 3.3.12. *If $\tau : F \Rightarrow G$ is a natural transformation of left (right) Quillen functors, then $\mathbf{L}\tau$ ($\mathbf{R}\tau$) is a natural isomorphism if and only if τ_x is a weak equivalence for all cofibrant (fibrant) x.*

Proof. Follows directly from definition of τ and theorem 1.2.18. ■

Remark 3.3.13. If we modify our notation to use $\mathrm{Ho}(F,U,\varphi)$ for the derived adjunction, and $\mathrm{Ho}\,\tau$ for the derived natural transformation, then we may restate some above results as noting that $\mathrm{Ho} : \mathcal{M}\mathrm{od} \to \mathcal{A}\mathrm{dj}$ is a pseudofunctor which commutes with the duality 2-functor D. We then note that a Quillen adjunction (F,U,φ) is a Quillen equivalence if and only if $D(F,U,\varphi)$ is.